Cross-Income Exposure in Schools and Long-Run Student Outcomes*

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Abstract

I examine how economic segregation in schools affects students' economic mobility using data from Texas. Lower-income (always on free/reduced lunch) students are exposed to nearly ten times fewer upper-income (never on free/reduced lunch) classmates than wealthier peers. To isolate peer effects, I exploit within-school, between-cohort variation in the share of upper-income students, controlling for school trends. Greater exposure to upper-income peers increases college enrollment and wages for lower-income students, driven by exposure to lower-achieving upper-income peers. Exposure to lower-income peers has no detectable impact on upper-income students' wages in early adulthood, suggesting cross-income exposure can improve equity with limited costs.

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1 Introduction

An important goal of education is economic mobility, but the likelihood of a U.S. child earning more than their parents has declined in recent decades (Chetty et al., 2017). One factor long theorized to relate to economic mobility is social capital, defined as relationships with others that enable individuals to access resources (Bourdieu, 1986; Coleman, 1988).[1] Most recent work suggests that friendships with higher-income individuals (cross-income social capital) account for a substantial portion of the variation in economic mobility across neighborhoods (Chetty et al., 2022). However, we know little about how institutional structures, such as schools, shape students' opportunities to form those relationships and why exposure to upper-income peers might matter for economic mobility.

To my knowledge, this is the first study to document the cumulative classroom exposure of lower-income students to higher-income peers throughout middle and high school. It is also the first to estimate the impact of peer family income in secondary school—specifically, the share of higher-income (never on free/reduced lunch) classmates on lower-income (always on free/reduced lunch) students' long-run outcomes, including college enrollment and early-career wages. The analysis leverages rich administrative data from Texas, a large, demographically representative U.S. state, which allows me to link student records—including student classroom assignment and test scores—to postsecondary and labor market outcomes. I find that lower-income students are exposed to significantly fewer higher-income classmates than their wealthier peers, largely due to residential and school-level socioeconomic segregation rather than within-school classroom assignment.[2] Finally, using within-school, between-cohort variation in the share of upper-income students, I show that lower-income students in cohorts with greater exposure to higher-income peers are more likely to enroll in college and earn higher wages in early adulthood. I also provide suggestive evidence that peer income may affect students through channels distinct from peer academic achievement.

I use "lower income" and "higher/upper income" as shorthand for students who are always below the free/reduced lunch income eligibility cutoff (income below $51,338 for a family of four) and

1. Social capital theory has been applied in education and employment to identify the drivers of inequities (e.g., Bourdieu, 1986; Coleman, 1988; Fernandez and Fernandez-Mateo, 2006; De Giorgi, Pellizzari, and Redaelli, 2010; Schmutte, 2015).

2. This finding aligns with Owens, Reardon, and Jencks (2016), who estimate that roughly two-thirds of income-based school segregation reflects segregation between school districts. However, their analysis did not include classroom-level data and thus could not assess the role of within-school sorting.

those who are always above the free/reduced lunch eligibility cutoff, respectively. Upper-income students constitute approximately 24% of students enrolled in public schools in Texas.[3] In the descriptive portion of this paper, I define exposure to upper-income students as the share of a student's cumulative number of classmates between grades 5 and 12 who are upper income. I find that the median lower-income students' share of upper-income classmates is 6% compared to 51% for the median upper-income student, and one in every five lower-income students goes through schooling with a share of upper-income classmates smaller than 1%.[4]

To address selection concerns and isolate the role of peers independent of access to resources, I exploit variation in peer composition across adjacent cohorts within the same school, controlling for school trend. Here, unlike in the descriptive portion of the paper, I use a sample of ten cohorts of high school students so that I am able to follow them into adulthood. The temporary (small) changes in a cohort's share of upper-income peers in a school are less likely to impact students' access to school resources and so can capture if there are peer family income spillovers that are independent of access to school resources (e.g., passage of institutional knowledge) that impact lower-income students' long-term outcomes. Consistent with the assumption that small within school deviations in the share of upper-income peers are unlikely to change access to school resources, I find that residual variation in peer income has no (detectable) impact on lower-income students' proportion of novice teachers, number of advanced courses offered, and average school per-pupil spending.

It is important to disentangle peer family income spillovers from access to observable school resources. If the positive relationship we observe between the share of upper-income peers and lower-income students' outcomes is driven only by differences in access to school resources (e.g., schools with more upper-income students tend to have more experienced teachers), then policies that equalize access to resources could improve outcomes without reducing economic segregation. If, however, the effects operate through peer income spillovers—independent of observable resource differences—then addressing school economic segregation would be necessary to improve outcomes for lower-income students.

3. The income cutoffs described are based on free/reduced lunch income eligibility in 2022. This definition of economic disadvantage builds on Michelmore and Dynarski (2017), who find that the number of years on free/reduced lunch captures student economic disadvantage better than a binary measure of economic disadvantage based on one year of free/reduced lunch eligibility status. Among students for whom financial aid data are available, I find that those with lower and higher incomes have a median parental income of approximately $23,033 and $131,294, respectively.

4. The median lower-income student goes through grades 5 to 12 having met 39 upper-income students out of around 700 (unique) students, compared to 325 upper-income students for the median upper-income student.

I find that a 2 percentage point, or one standard deviation (SD), increase in the share of upper-income peers increases lower-income students' likelihood of enrolling in college by about 0.11 percentage points (0.3%) and raises their early adult annual wage by $87 (0.4%). The observed increase in annual wages is around three times the estimated return to four-year college enrollment based on Zimmerman (2014), suggesting that the impact on wages likely does not operate exclusively through the increase in college enrollment.[5] The positive impact of marginal, temporary changes in the share of upper-income students on lower-income students' wages suggests that resource equalization policies alone may be insufficient to address the harms of economic segregation.

The impact of the share of upper-income peers seems to be driven by peer income and not other characteristics correlated with income like race, achievement or disciplinary record. I show that alternative peer composition measures, such as the share of White peers and the share of peers with prior disciplinary records, have no detectable effects on lower-income students' college enrollment or wages. The estimated impact of upper-income peers is consistent when controlling for the share of White peers, peer achievement and peer disciplinary record.

I show that the positive impact of having a higher share of upper-income students appears to be driven by upper-income lower-achieving students. That the positive effect of upper-income peers is driven by upper-income, lower-achieving students challenges the common emphasis on peer academic spillovers as the primary mechanism in the education production function and suggests a broader role for other potential social capital mechanisms: the transmission of institutional knowledge, norm-setting, or enhanced aspirations through economically advantaged peers. I find suggestive evidence consistent with a "proximity" mechanism: upper-income peers appear more beneficial for lower-income students when they are closer in achievement and are of the same racial/ethnic group.[6] Prior work on peer effects in k-12 has generally focused on peer academic achievement (see, e.g., Imberman et al, 2012; Lavy, Paserman and Schlosser, 2012; Jackson, 2013; Feld and Zoelitz, 2017).

Under strong extrapolation assumptions (e.g., the marginal effects identified from small, within-

5. The estimated wage increase is ten times the estimated wage gain from Mountjoy (2022) for inducing students to go from no college to two-year college enrollment. The estimated increase in college enrollment is mainly driven by more lower-income students enrolling in two-year then four-year public colleges.

6. While I can rule out some alternative mechanisms correlated with peer income, the design does not allow me to distinguish which specific channel drives the effects.

school cohort shocks scale approximately linearly to larger changes in peer composition and that moving students across schools or districts does not introduce offsetting changes in other inputs or selection), I use the estimates to project the potential gains from reducing economic segregation within and across districts. I also construct bounds using the larger, though potentially more confounded, between-school variation in upper-income share; these bounds suggest that estimates based on within-school cohort deviations are similar to those implied by the between-school variation. The resulting projections imply that eliminating within-district segregation across schools would raise lower-income students' wages by about 0.5% to 0.7%, while eliminating both within- and between-district segregation would raise average lower-income students' wages by about 2.2% to 2.8%.

Lastly, I test whether increased exposure to lower-income peers adversely affects upper-income students. I find that a 2 p.p. (1 SD) increase in the share of lower-income peers (holding constant the proportion of middle-income peers, i.e. swapping upper-income students with lower-income peers) slightly decreases upper-income students' likelihood of graduating from college (-0.3% (p = 0.05)), though the small decrease in graduation rate does not seem to be reflected in upper-income students' wages—I find no detectable impact on upper-income students wages (-0.2% (p = 0.3)). The small null impact on wages highlights the possibility that policies designed to improve cross-income exposure could be efficiency-enhancing——or at least neutral——helping lower-income students without imposing significant costs on their higher-income peers.

I complement my administrative data analysis with high school-level measures of cross-income friendship networks from Chetty et al. (2022) and student-level friendship data from the Add Health study. These comparisons show that school-level exposure to upper-income peers is highly predictive of both school-level economic connectedness and friending bias, as well as individual-level friendship formation across income groups.[7]

This paper makes three main contributions. First, to my knowledge it is the first to quantify how economic segregation across districts, schools, and classrooms generates large differences in students' cumulative exposure to upper-income classmates, using longitudinal classroom rosters.

7. The correlation between exposure to upper-income classmates as measured in this paper and economic connectedness as measured by Chetty et al. (2022) is 0.9. Friending bias as measured by Chetty et al. (2022) is the likelihood of befriending high-SES students conditional on school composition. I find that friending bias in schools is highly correlated with the level of sorting by income across classrooms in a school: the correlation is approximately 0.6.

Relative to prior work, I also use a richer administrative proxy for family income—students' *years* eligible for free/reduced-price lunch, rather than a single-year indicator. Existing segregation research has focused primarily on racial/ethnic segregation (e.g., Clotfelter, Ladd, and Vigdor, 2002; Clotfelter, Ladd, Clifton and Turaeva, 2021) and within-school academic sorting (e.g. Lucas and Berends, 2002; Antonovics, Black, Cullen, and Meiselman, 2022). The smaller literature on economic segregation is often constrained by data that lack classroom-level observations and/or links to longer-run outcomes such as college enrollment and earnings (Owens, Reardon, and Jencks, 2016; Kalogrides and Loeb, 2013; Dalane and Marcotte, 2022).

Spiegel et al. (2025) is a notable exception, using family income derived from IRS records to document cross-income exposure, but it is limited to Oregon, a context with different demographics and relatively lower levels of school choice. They also do not link these exposure patterns to student outcomes. Finally, I show that classroom-based measures of cross-income exposure constructed from administrative data closely proxy cross-income social capital as measured using Facebook friendship links.

Second, it's the first paper to my knowledge to try to isolate the role of peer income spillovers in secondary school on lower-income students' college enrollment and wages using idiosyncratic variation in the share of upper-income peers, and its relation to peer achievement spillovers. I contribute to the large peer effects literature examining the relationship between peer composition and short- and long-term outcomes (examples include Sacerdote, 2001, 2011; Zimmerman, 2003; Lavy, Paserman and Schlosser, 2011; Black, Devereux and Salvanes, 2013; Cools, Fernandez and Patacchini, 2019; Zimmerman, 2019; and Rao, 2019; Carrell, Hoekstra and Kuka, 2018).

This paper is most closely related to work by Cattan, Salvanes and Tominey (2025), which investigates the impact of exposure to the children of parents who attended elite schools in Norway. I extend Cattan, Salvanes and Tominey (2025) work in two key ways. First, I quantify how common cross-income exposure is in middle and high school classrooms and decompose how much of that exposure arises from segregation at the district, school, and classroom levels. Second, I identify the impact of exposure to upper-income peers on students' wages—effects that may not necessarily operate through parents' education and that Cattan, Salvanes, and Tominey (2025) were not able to capture. I show that the wage effects are larger than what would be predicted based on the impact on college enrollment alone, suggesting that peer income spillovers may operate through additional mechanisms beyond higher education pathways.

The paper is organized as follows. In Section 2, I begin by laying out the theoretical framework of the analysis. Section 3 defines my measures of cross-income exposure and data used. Section 4 reports estimates of cross-income exposure at the state level and how they compare to cross-test score exposure, as well as district, school and classroom integration benchmarks, as well as describes the relationship between exposure to upper-income students and friendship formation. Section 5 reports the relationship between cross-income exposure and long-term student outcomes. Section 6 concludes the paper.

2 Conceptual Framework

Exposure to upper-income peers is a function of the district a student resides in, the school they enroll in, and the classroom they are assigned to or choose. At each of these levels, parents and students make decisions about where to live and what school and classroom to enroll in given their economic resources (household income) and subject to potential information imperfections and discriminatory barriers. These decisions and constraints shape not only their exposure to upper-income peers but also their access to neighborhood and school resources. The first part of the paper attempts to draw a complete picture of how each of these layers (district, school, and classroom) contributes to lower-income students' exposure to upper-income peers across schooling years.

Theoretically, exposure to upper-income peers could shape lower-income students' long-term outcomes in at least two ways: resource allocation and/or social capital.[8] The distribution of students by income across and within schools could shape how financial and instructional resources are allocated (Owens and Candipan, 2019 and Kalogrides and Loeb, 2013). Owens and Candipan (2019) find that schools in upper-income neighborhoods tend to have higher spending per student and higher salaries for teachers. Unequal resource allocation can also take place within a school, between classrooms. Kalogrides and Loeb (2013) find that upper-income students tend to be assigned more experienced teachers than lower-income students in the same school. I find a similar

8. Other potential channels include classroom spillover and class rank (Denning, Murphy and Weinhardt, 2023).

pattern in Texas. Students with 10 pp more upper-income classmates, within the same school, are in classrooms with 3 pp fewer novice teachers (with three or less years of experience), on average.[9]

Another potential channel through which exposure to upper-income peers might matter is social capital. Social capital may impact students' long-term outcomes through information and/or norm setting (Coleman, 1988). An example of an information channel is if higher-income students are more likely to know which courses to take to improve their likelihood of being accepted into more selective colleges and that they are more likely to share this information with lower-income students when sharing the same classroom.[10] The social norms channel depends on the prevalence (actual/assumed) of a behavior in a group. For example, if upper-income students are more likely to enroll in selective universities, this may create a norm/expectation that students apply and enroll in selective universities across income groups.[11]

The strength of the information and norm channels may depend on the school and social structure. If upper- and lower-income students are likely to attend different classrooms (because of high classroom sorting by income/test score), then they may be less likely to pass on information and shape cross-group norms.[12] It may also depends on the strength of the cross-group relationships. If students are less likely to form friendships across income categories, then they may be less affected by cross-income exposure. Hoxby (2001) finds that peer achievement effects are stronger within a race. Cattan, Salvanes, and Tominey (2025) find that the impact of classmates whose parents graduated from elite schools is stronger among high-SES students than among low-SES students, suggesting that peer effects may be weaker across income groups. I present evidence in section 5.4 suggesting that exposure to upper-income peers of the same race/ethnicity and lower-achieving upper-income peers may have a larger impact on lower-income students.

Alternatively, the social capital channel may operate exclusively through students' academic achievement. In other words, it could be that upper-income students happen to perform better

9. The raw relationship between the share of upper-income classmates and access to school resources is summarized in Figures A10 and A11 in the Supplementary Appendix. The relationship between advanced course taking and exposure to upper-income students is more ambiguous, although generally positively correlated with exposure to upper-income students. In contrast, school spending per pupil is negatively correlated with higher exposure to upper-income students.

10. Bourdieu (1986) suggests that individuals from upper-income households have higher cultural capital (knowledge, skills, and tastes belonging to a social class), which results in the persistence and reproduction of social structures.

11. Papers that documented the potential importance of social norms on academic performance in schools, perceptions, and giving behavior include Frey and Meier (2004), Bursztyn and Jensen (2015), Bursztyn, González, and Yanagizawa-Drott (2020).

12. I find that changes in cohort income composition are not perfectly reflected in changes in lower-income students' classroom peer composition, suggesting that sorting within a school across classrooms may reduce the impact of increases in a schools' cohort income composition.

academically, and being around higher achieving students–independent of their income–has an impact on lower-income students' academic performance and labor outcomes. Some of the channels through which a higher proportion of higher achieving peers may impact student outcomes include: student class rank, teacher behavior and/or academic spillover through peer-to-peer tutoring. The peer effects literature suggests the impact of peer achievement may vary depending on students' own baseline academic performance (Cools, Fernández and Patacchini, 2019; Feld and Zölitz, 2017, Lavy, Paserman and Schlosser, 2011; Sacerdot, 2011).

The type of variation I use in the causal analysis (Section 5)—small, temporary deviations from a school's average peer composition—may not be large enough to alter classroom norms or school resources. However, such variation may affect the amount of information available to lower-income students, shaping their college choices and employment outcomes. In this paper, I focus on changes in the share of upper-income peers, which do not necessarily translate directly into cross-income relationships (social capital). That said, in Section 4.3 I show that the share of upper-income peers in a school is strongly predictive of the likelihood of forming cross-income friendships.

3 Data and Measures of Cross-Income Exposure

3.1 Data

I use longitudinal administrative data from the Texas Education Research Center (ERC) that link student data from the Texas Education Agency (TEA) to Texas Higher Education Coordination Board (THECB) and Texas Workforce Commission (TWC) data. These TEA data cover the period from 2004 to 2022 and include student test scores, courses (and class assignment starting in 2011), demographics, attendance, graduation, and assigned teachers (including teacher certification and demographics). My main measure of income is free/reduced lunch status. I use years of assignment to free/reduced lunch status to capture the degree of economic disadvantage.[13]

I use college enrollment data (THECB) from 2014 to 2023 data. I focus on the following college enrollment outcomes: any college enrollment, two-year public college enrollment, four-year public college enrollment, highly selective college enrollment, and graduation from any college. Any college

13. Test scores are mainly based on standardized grade 4 and 8 TAKS (2007–2011) and STAAR (2012–2018) reading and math tests.

enrollment includes public colleges (two-year and four-year), career and technical education (up to 2020), health colleges and private colleges.[14] For public two-year, public four-year and selective college enrollment, the categories are not mutually exclusive. For example, a student who enrolls first in a two-year college and later in a four-year college is counted in both the two-year and four-year enrollment outcomes. I also use financial aid application data to identify parental income and if a student applies for financial aid. The college data are limited to enrollments in the state of Texas. The employment data (TWC) includes information on Quarterly wage. The employment data are based on unemployment insurance data and hence cover only employment in Texas. In Section 5, I use quarterly wage data from 2023 (the most recent employment data).

To identify and better understand the link between exposure and friendship formation, in Section 5, I supplement the analysis with two other sources of data: Add Health data and publicly available high school measures of the likelihood of cross-income friendship based on Facebook data from Chetty et al. (2022), both of which I discuss in more detail in Section 5.

3.2 Measure of Family Income and Cross-Income Exposure

To measure student income, ideally, I would have parental income for every enrolled student. In the absence of data on parental income, I use the proportion of years on free/reduced lunch to capture the degree of economic disadvantage.[15] I divide students into three main income groups: always, sometimes and never on free/reduced lunch. I define upper-income students as those never in free/reduced lunch status. Upper-income students represent approximately 24% of the student population. Students always in free/reduced lunch status (lower-income students) constitute approximately 29% of the student population.

The segmentation of students by the proportion of years eligible for free or reduced-price lunch appears to capture meaningful variation in parental income, as shown in Figure 1. Among students with financial aid data, the median reported parental income is $131,294 for those never eligible, $42,879 for those sometimes eligible, and $23,033 for those always eligible.[16] Because the financial

14. Colleges are defined as selective if they are between levels 1 and 4 based on Barron's selectivity index (most to very competitive in 2009). Public 2-year and 4-years make the majority of enrollment in 2020 (85%). Private colleges make up 8%.

15. To identify the proportion of years on free/reduced lunch, I use all years from 2004 to 2022.

16. The median upper-income student in a lower-income student's classroom may have a lower income than the median upper-income student overall. To assess this, I calculate the median reported FAD income among higher-income (never on FRPL) students, weighting each student by their share of lower-income (always on FRPL) classmates. The resulting median income is

aid data reflect parental income at a single point in time (the end of secondary school, just before college entry), they may be sensitive to temporary income shocks and may not represent permanent or long-term parental income. Students with financial aid data are also a select, systematically different, group.[17] Nevertheless, it can provide us with a general sense of the variation.[18]

For the rest of the paper, I refer to students who are never on free/reduced lunch as higher- or upper-income. I refer to students who are sometimes and always on free/reduced lunch as middle- and lower-income students, respectively. This paper focuses on lower-income (always on FRPL) students' exposure to upper-income (never on FRPL) peers and how it relates to their long-term outcomes.

I define cross-income exposure as the share of classmates of another income group in a student's classroom, focusing on the share of upper-income students. I capture the cumulative exposure to upper-income students by following three cohorts of students (2012–2014) starting from fifth to expected twelfth grade in Texas. In each year, I capture the number of classmates in a student's classroom (excluding self). I also capture the number of upper-income classmates in the classroom (excluding own status). I divide the total number of upper-income peers student i encountered in all the classrooms they were enrolled in between grades five and twelve ($N^H_{(-i)}$) by the total number of peers student i encountered in those classrooms ($N_{(-i)}$): $\dfrac{N^H_{(-i)}}{N_{(-i)}}$.

One can understand this measure as capturing the proportion of potential interactions rather than the proportion of students. If a student is in multiple classrooms with the same student, the latter is included in the denominator (total student interactions) multiple times. She would also be included in the numerator if she were of upper income multiple times. The assumption is that what matters is the proportion of interactions rather than the proportion of students. Each interaction is likely to yield an additional benefit. I assume that each interaction has an equal level of benefit.

only slightly lower: $115,016 (vs. $131,294).

17. Financial aid data are available for 51%, 40% and 34% of students who are never, sometimes, and always in free/reduced lunch status, respectively.

18. For students who apply to public universities in Texas and report parents' income bracket, 81% 24% and 3% who were never, sometimes and always on free/reduced lunch, respectively, report having parental income above $80,000.

4 Descriptive Patterns: How Likely is Cross-Income Exposure and What is the Contribution of District, School and Classroom Economic Segregation?

4.1 Sample

To capture students' exposure to upper-income classmates, I follow three cohorts of students from grade 5 to expected grade 12. The three cohorts are the 2012, 2013 and 2014 cohorts. Students enrolled in any public school (including charter schools) in Texas in grade 5 in academic year 2012 belong to the 2012 cohort. These cohorts of students are followed in the data, and their classroom income composition is documented every year from 2012 to 2019. Across these cohorts, there are 1,128,554 students. Table 1 summarizes the characteristics of the sample. Approximately 50% of the cohort identify as Latin-American students, 14% as Black students, and 68% as White students. Most of the students are enrolled in traditional public schools—4–6%, in a given grade, are enrolled in a charter school, as shown in Table A1. Note that "expected" grade is not their actual grade but the grade in which we expect them to be enrolled given the year and their cohort.

4.2 Descriptive Patterns of Exposure to Upper-Income Students

Lower-income students go through school sharing very few classrooms with upper-income classmates. The average lower-income student goes through grades five to twelve with 10% of their classmates upper-income, compared to 51% for upper-income students (Figure 2). A large share of lower-income students are exposed to very few upper-income peers—the distribution of exposure is right-skewed (Figure 3, Panel (a)). The median lower-income student is in classrooms with 6% upper-income classmates, and for one in five lower-income students only 1% of classmates are upper-income. The median lower-income student goes through grades 5 to 12 having met 39 upper-income students out of roughly 700 students, compared to 325 for the median upper-income student.[19]

These patterns are not explained by racial and ethnic segregation. Lower-income students in every racial/ethnic group are exposed to fewer upper-income students than their upper-income

19. This statistic is based on a random sample of 3,000 students (1,000 from each cohort) where I count only the first interaction (first shared classroom) with a given upper-income student. The exposure gap based on first interactions is similar at 39 p.p.. Lower-income students in urban districts are exposed to the smallest share of upper-income students (8%), and the gap in exposure is largest in suburbs (Figure A1).

counterparts (Figure A2).[20] I also find gaps in exposure to same-race/ethnicity upper-income classmates within racial/ethnic groups (Figure A3), though the gaps are smaller.[21]

The finding that lower-income students are exposed to relatively few upper-income classmates is robust to alternative income definitions. Using financial aid records, the average lower-income student goes through school with 13% of classmates reporting parental income in the top 24 percentiles, compared to 43% for upper-income students (Figure A4).[22]

The difference in exposure to upper-income students reflects residential, school, and classroom socioeconomic segregation. To separate these components, I compare students' exposure to three benchmarks. In the first, students are randomly assigned across districts, schools, and classrooms (district integration benchmark). In the second, district composition is fixed and students are randomly assigned to schools and classrooms (school integration benchmark). In the third, school composition is fixed and students are randomly assigned to classrooms (classroom integration benchmark). In practice, this is similar to comparing lower-income students' exposure to the share of upper-income students in the state, district, and school.[23] Figure 2 plots these benchmarks alongside actual exposure.

Total income segregation is the gap between actual exposure and the district integration benchmark: 11.6 percentage points (Figure 2, Panel (a)). In other words, under random assignment across districts, lower-income students would be in classrooms with 11.6 percentage points more upper-income classmates—more than twice their current exposure. Seventy percent of total income segregation is attributable to across-district sorting, 23% to across-school sorting within districts, and 6% to across-classroom sorting within schools.[24] The relatively large district component is consistent with Owens, Reardon, and Jencks (2016), who find that roughly two-thirds of income segregation between schools reflects segregation between districts. Spiegel et al. (2025) similarly find, using a different measure of family income, IRS records, that classrooms account for a small

20. White students, across incomes, are exposed to about 10 pp more upper-income classmates.

21. The low exposure of Hispanic and Black lower-income students to upper-income classmates of the same race/ethnicity is concerning if peer effects are stronger within racial groups.

22. Financial aid data may understate the gap due to left- and right-tail missingness. Based on these data, classmates' average parental income is $47,152 for lower-income students and $108,859 for upper-income students.

23. Because I exclude students' own status, expected exposure under random assignment differs slightly by income group, especially in small units such as classrooms. Supplementary Appendix A details these calculations.

24. Each share is computed as the difference between adjacent benchmarks divided by total income segregation.

share.[25]

The gap in exposure is much smaller under the traditional contemporaneous FRPL definition (FRPL vs. non-FRPL), as used in Reardon, Owens, and Jencks (2016), than under my multi-year FRPL measure (Figure 2, Panel (b)). Under contemporaneous FRPL, FRPL students' exposure to non-FRPL classmates is about one-third the exposure experienced by non-FRPL students. The distribution is also less right-skewed, with the median close to the mean (about 27% non-FRPL classmates). Despite level differences, the decomposition of total income segregation is similar: 70% across districts, 25% across schools within districts, and 4% across classrooms within schools.[26]

The difference in exposure to higher-achieving peers is considerably smaller than the difference in exposure to upper-income peers. I define higher-achieving students as those scoring in the top 24 percentiles on grade 4 reading, matching the upper-income share (24%).[27] As shown in Figure 4, lower-income students are exposed to 20 percentage points fewer high-achieving peers in grades 5–12 than upper-income students. In comparison, the gap in exposure to upper-income students is closer to 41 percentage points.[28]

Students are also more clustered by income than by test score. The distribution of exposure to upper-income peers is right-skewed, while exposure to high-achieving peers is closer to normal (Figure 3 Panel (b)). At the 25th percentile of the peer income distribution, lower-income students are exposed to 1.4% upper-income classmates; at the 25th percentile of the peer achievement distribution, they are exposed to 12% high-achieving classmates.[29]

4.3 How Does Cross-Income Exposure Relate to Friendship Formation?

Exposure to upper-income (never on free/reduced lunch) peers as captured in this paper is highly correlated with a school's expected rate of cross-income friendship formation (economic connectedness) as captured by Chetty et al. (2022). I link the Texas administrative data with public

25. While classroom assignment explains only a small portion of the average gap, classroom-level policies may be easier to implement; in contrast, district-level policies such as charter expansion appear to have even smaller effects on cross-group exposure (Monarrez, Kisida, and Chingos, 2022). Classroom integration is also more important in higher-income districts (Figure A5).

26. The classroom share is slightly smaller (4% rather than 6%) when using contemporaneous FRPL.

27. Results are similar using grade 4 math.

28. The gap in cross-test score exposure is smaller than cross-income exposure (Figure A6). Low-achieving students are those in the bottom 29 percentiles of grade 4 reading.

29. Test scores may be measured with more error than income, which is based on multiple years of FRPL status. I find similar patterns if I define achievement by students' average grades 3 to 8 test scores instead as shown in Figure A7.

high school measures on "economic connectedness" captured by Chetty et al., (2022). Economic connectedness is defined as the share of high-SES friends among low-SES individuals on Facebook divided by the share of high-SES individuals in the population. For each high school, I construct an exposure measure equal to the average share of upper-income classmates among lower-income students in 2012.[30]

I find that school-level exposure to upper-income classmates is strongly predictive of economic connectedness (Figure 5, Panel (a)). The correlation between a school's exposure measure and economic connectedness ranges from 0.86 to 0.93.[31] The strong relationship between this paper's measure of exposure to upper-income classmates and economic connectedness as measured by Chetty et al. (2022) can be viewed as a validity check: both measures appear to capture the extent of cross-income interaction within a school.

Within-school income sorting is also strongly related to cross-income friendships, conditional on school composition, as captured by Chetty et al. (2022)'s "friending bias" measure.[32] The correlation between the variance ratio and friending bias is 0.61 to 0.67 (Figure 5, Panel (b)). The strong correlation between school classroom sorting by income and friending bias suggests that friending bias captures, at least in part, differences across schools in lower-income students' classroom-level exposure to upper-income peers, conditional on school composition.

To examine the relationship between an individual student's likelihood of forming friendships with upper-income peers and their school's exposure to upper-income students I use the Add Health data on close friendships (students' top five friends). I find a student's school and extracurricular composition (cross-income exposure) predicts the SES composition of their close-friend network, though more modestly: the correlations are 0.37 for school composition and 0.40 for extracurricular composition (Figure A8).[33]

30. I use 2012 rather than 2019 because the friendship measures are based on Facebook networks for individuals ages 25–44 as of 2022. I merge 1,132 Texas schools to the high school friendship data (51% of schools serving grades 9–12); 981 of these have economic connectedness measures.

31. The range is based on whether I use the economic connectedness measure based on students' own income or students' parental income.

32. I measure within-school sorting by income using the variance ratio: the difference between lower- and upper-income students' average share of upper-income classmates within a school.

33. The fitted line lies below the 45-degree line, indicating that exposure to high-SES students does not translate one-for-one into high-SES close friendships, though it is close. For low-SES students, the slope is 0.78 for school composition and 0.71 for extracurricular composition. See Supplementary Appendix B for how I defined SES in the Add Health data.

5 How Does Cross-Income Exposure in Schools Affect Lower-Income Students' College Enrollment and Employment?

5.1 Sample

Up to this point the goal was to understand how likely cross-income exposure in schools is and how it relates to friendship formation. In order to understand students' schooling experience, I followed three cohorts of students, starting with the earliest cohort we can observe classroom data for—the 2012 cohort—to the last cohort we can follow from grade 5 up to expected grade 12 (i.e., up to 2019, before COVID)–the 2014 cohort.

In this section, I examine if and why the share of upper-income classmates in a school matters for lower-income students' college enrollment and employment outcomes. To improve precision given the small variation year-to-year in the share of upper-income students in a school, I need to follow a large number of cohorts in a school into adulthood, therefore the sample in this section is different. I identify ten cohorts of grade 9 students starting in 2010. Students are grouped into cohorts based on the year they are enrolled in grade 9. For example, students enrolled in grade 9 in 2009–2010 are grouped into the 2010 cohort.[34]

I observe ten cohorts of students (2010 to 2019) for the college enrollment and secondary school outcomes. For the college graduation and wages in early adulthood outcomes, I follow five cohorts of students (2010 to 2014), to allow for enough time for students to graduate college and find employment at expected age 23–28.

5.2 Identification Strategy

Descriptively, I find that students independent of grade 8 reading (baseline) test scores appear to perform better when in classrooms with more upper-income classmates during their high school years, as shown in Figure A9. The positive relationship between a student's proportion of upper-income classmates and long-term outcomes may be driven by differences in access to school resources

34. The income measure is based on the number of years on free/reduced price lunch. I observe each cohort for 10 years. For example, students in the 2010 cohort who never had free/reduced lunch between 2004 and 2013 are identified as higher income. The data are at the student-school level. Students are assigned to their grade 9 school. Some students are enrolled in multiple schools in grade 9.

correlated with the proportion of upper-income classmates (Figure A10 and Figure A11).

One way the peer effects literature has gone around identifying the relationship between peer composition and individual student outcomes is by using (small) changes in cohort composition between adjacent cohorts in the same school. These marginal, temporary, changes in cohort composition are less likely to change students' access to school resources and to be predicted by parents choosing to enroll their kids in a school. The assumption is that adjacent cohorts of students who attend the same school would have a similar school environment and the only difference between them is that some happen to have a slightly different cohort composition (e.g., more upper-income students) due to random draws from the school population that are unrelated to changes in school characteristics. Using cross-cohort deviations in peer composition to capture peer effects dates back to Hoxby (2001) and since then has been used extensively in the peer effects literature.[35]

Building on Hoxby (2001), I use within-school between-cohort deviations in the share of upper-income peers to capture the impact of having more upper-income peers on lower-income students' outcomes. Note that in this section (unlike in prior sections) I use variation in grade 9 school cohort composition regardless of whether those students end up in the same classroom.[36] My preferred specification is shown in Equation (1):

$$Y_{isc} = \beta_0 + \beta_1 PropHighIncome_{-isc} + X'_{isc}\beta_2 + \delta_c + \delta_s + c \times \delta_s + \epsilon_{isc} \tag{1}$$

where $PropHighIncome_{-isc}$ is the share of student i's peers (excluding herself) in cohort c that are upper income; β_1 captures the impact of an unexpected change (residual of the linear school trend) in the composition of peers on the long-term outcome; and X'_{isc} is a vector of student i's characteristics (gender, race/ethnicity and math and reading grade 8 test scores). δ_s captures a school fixed effect to account for level school differences in student outcomes, and $c \times \delta_s$ captures school-specific time trends (c is a linear cohort trend). I also include cohort fixed effects, δ_c, to capture any cohort-specific changes in the sample. Because the treatment is at the school-cohort level, I cluster the standard errors at the school level. I run the regression only for lower-income students since I am interested in how exposure to upper-income peers impacts lower-income

35. For examples see Feld and Zölitz (2017), Lavy and Schlosser (2011) and Angrist and Lang (2004).

36. I use variation in school-cohort instead of classroom-cohort because I cannot track the same classrooms over time and students self select into classrooms.

students' long-term outcomes.

Identifying Variation. In Equation (1), the identifying variation comes from deviations in a cohort's share of upper-income students from the school-specific trend. The residual standard deviation is 2 percentage points.[37] Cohorts at the 1st percentile have 5.9 percentage points fewer upper-income students than predicted by the trend, while cohorts at the 99th percentile have 6.2 percentage points more (Table 2 and Figure A12).

The median lower-income student is in a school-cohort of 578 students, 8% of whom are upper-income. For median lower-income student, a 2 percentage point increase in the upper-income share implies an increase from 46 to 58 upper-income peers. Under random classroom assignment, this corresponds to an increase from 1.4 to 1.7 upper-income students in a median class of 17.[38]

These temporary, small shocks to cohort composition are unlikely to alter school-level resources (e.g., teacher quality, AP offerings, or budgets) and are also unlikely to shift school-wide norms. Instead, they may operate through more localized interactions, such as the diffusion of institutional knowledge or changes in students' beliefs about college options, or through teacher behavior and expectations that respond to small changes in classroom composition.[39]

Validity of the Identification Strategy. The identifying assumption is that conditional on school-specific time trends, deviations in a cohort's share of upper-income students are as good as random—that is, they are uncorrelated with other cohort-level shocks that affect lower-income students' outcomes. Such deviations could arise from idiosyncratic demographic fluctuations within a school's catchment area that are unrelated to lower-income students' potential outcomes (e.g., a small number of upper-income families with children happen to have kids in particular birth cohorts increasing the number of upper-income students in those cohorts slightly relative to others).

The assumption that the fluctuations in the share of upper-income students are random would be violated if parents observe and respond to cohort-to-cohort composition changes (e.g., by moving or switching schools), or if schools experience contemporaneous changes that affect both outcomes

37. The residual standard deviation is similar to the median standard deviation expected from random draws from the school population (Figure 6).

38. There is high variation in school-cohort sizes: 10% of lower-income students are in school-cohorts with 100 or fewer students. There is also a lot of variation in the share of upper-income students: 10% of lower-income students are in school cohorts with close to 0% upper-income peers.

39. Sections 5.4 and 5.5 provide suggestive evidence on potential mechanisms including peer achievement, peer racial composition and observable school resources.

and cohort composition (e.g., the introduction of advanced coursework). To address these concerns, Equation (1) includes school-specific linear time trends, which capture potentially slow-moving evolution of neighborhood composition and school characteristics. Identifying variation therefore comes from "unexpected" (temporary) deviations from these trends, as in other peer-effects designs that rely on within-school cohort variation around school trends (Hoxby, 2000; Lavy and Schlosser, 2011; Lavy, Paserman and Schlosser, 2012; Getik and Meier, 2025).

We might be concerned that the deviations in the share of upper-income students from the school linear time trend may be capturing other simultaneous changes in lower-income students' characteristics that are correlated with their college enrollment and wages (common shocks). For example, cohorts with unusually high upper-income shares might also include lower-income students with higher household income, mechanically raising college enrollment. I assess the possibility that lower-income students in cohorts with higher deviations in the share of upper-income peers look any different using balance tests that relate deviations from the school trend in upper-income share to predetermined demographic characteristics of lower-income students, including a summary index of predicted college enrollment and wages based on student demographics. Conditional on school trends, deviations in upper-income students' share are not systematically related to lower-income students' baseline characteristics (gender, race, and immigrant status), as shown in Table 3, Model (3)–the coefficients are small and insignificant.[40] Lower-income students' parental income reported on financial aid data also appears to be no different when exposed to more upper-income peers, suggesting that the estimates are not driven by cohort specific changes in lower-income students' household income that correlate with better long-term outcomes.[41] By contrast, when school trends are omitted there is evidence of imbalance (Table 3, Models (1) and (2)), underscoring the importance of conditioning on school trends.

A remaining concern is that some schools' composition trends may be nonlinear. In that case, deviations from a linear trend could capture systematic (nonlinear) changes correlated with unobservables. I address this concern in several ways. First, I assess whether the magnitude of

40. Another concern is that when conditioning on peer achievement to identify if the effect is driven by peer achievement, deviations in students peer income may no longer be random. In Table 3, Model (4), I additionally control for peer achievement and continue to find no evidence of imbalance in lower-income students' demographics. The models in Table 3 replicate the models in the main results Tables, but without controlling for student demographics or test score. Appendix Table A3 reports balance tables that also include test score controls and demographic student characteristics (excluding the outcome).

41. Note that the financial aid data is based on a select group of students and the proportion of lower-income students applying for financial aid slightly increases (by 0.1 pp per 2 pp (one-SD) increase in upper-income share).

within-school cohort deviations resembles what would be expected from random sampling variation given each school's cohort size and average upper-income share. Figure 6 reports a Monte Carlo exercise: for each school, I simulate the number of upper-income students from a binomial distribution with probability equal to the school's average upper-income share, then compute the within-school standard deviation of residual variation, and repeat the simulation 1,000 times to form an empirical confidence interval. The expected size of the deviations also depends on the cohort size with larger cohort sizes expected to have smaller deviations.[42] The observed distribution of within-school residual variation is similar to the simulated distribution (Figure 6). Overall, 92% of schools fall within their simulated 95% confidence interval, close to what would be expected under random draws.[43] Re-estimating the main specifications restricting to schools whose residual standard deviation lies within the school-specific 95% interval yields similar estimates (Tables A6 and A7, Model (1)).

Second, I show that results are robust to alternative trend specifications and exclusion rules (Tables A6 and A7 Models (2)–(5). I (i) repeat the simulation exercise without linear trends and exclude schools whose deviations from the school mean that fall outside the simulated 95% intervals (Models (2)); (ii) allow for quadratic school trends (Model (3)); and (iii) bin the school-cohorts (assuming no systematic changes in adjacent cohorts within the bins) for the college enrollment outcomes: I bin the first and second five years in Model (4), and bin the first six years into two three-year bin and the last four years into one bin in Model (5).[44] Across specifications, the estimates remain positive and of similar magnitude, suggesting that results are not driven by the linear functional-form assumption. Third, I implement placebo (falsification) tests that replace a student's actual cohort composition with the composition of the adjacent cohort ($t-1$ or $t+1$) and show that the estimated effect is driven by lower-income students' *own* cohort share of upper-income peers.[45]

One question we might have is how much overlap there is between the share of upper-income students a student has in grade 9 and that in earlier grades. The treatment—deviations from

42. This approach follows Lavy and Schlosser's (2011) simulation-based check using gender composition. This method takes into account differences in the average share of upper-income students and school size.

43. The observed right tail is somewhat longer than the simulated distribution, in part because the simulation distribution uses averages across repeated draws.

44. I do not do bin the school-cohorts for the graduation and wage outcomes since they are based on five cohorts only.

45. Lavy and Schlosser (2011) use an analogous placebo test to detect time-varying school shocks.

high school trends in the proportion of upper-income students—might be capturing changes in the cohort's share of upper-income peers that date back to earlier school years.[46] My main specification controls for students' own grade 8 reading and math test score and so in so far as students' grade 8 test scores are capturing the impact of prior exposure to upper-income students, the remaining impact of exposure to upper-income students is capturing the impact of high school exposure. I also present estimates including fixed effects for students' grade 8 school-cohort to capture the impact of exposure to upper-income students in high school independent of middle school exposure and to address the concern that grade 8 test score may be a function of grade 8 exposure to upper-income peers. By including grade 8 school-cohort fixed effects I limit the variation to students who attended the same grade 8 school and cohort but happen to attend high schools with differing deviations from school trend in the proportion of upper-income students. The coefficients on the proportion of upper-income peers are generally consistent but less precise when controlling for grade 8 school-cohort as shown in Table A8 Model (5). The consistent estimates suggest that most of the impact captured in the main specification is driven by differences in high school exposure to upper-income peers, rather than earlier school years. We may be interested in the cumulative impact of deviations from school trends in the proportion of upper-income students on lower-income students. As such, I also present results without controlling for grade 8 test scores as shown in Table A8 Model (3)–the estimates are consistent, but slightly larger.

5.3 Main Results

I find evidence consistent with the idea that lower-income (always on FRPL) students benefit from being in cohorts with a higher share of upper-income (never on FRPL) peers. Using Equation (1), Table 4 shows that lower-income students' college enrollment increases when they are exposed to more upper-income peers, both with and without controlling for school-specific trends.[47]

In the preferred specification with school-specific trends (Table 4, Model (3)), the coefficient on the change in the share of upper-income peers on any college enrollment is 0.056. This implies that

46. I find that a 10 pp increase in the proportion of upper-income students in high school is associated with a 6 pp higher share of upper-income students in grade 8.

47. A higher share of upper-income peers is associated with both a decrease in the share of middle-income (sometimes on FRPL) and lower-income (always on FRPL) students—a 10 pp increase in the share of upper-income students is associated with a 5.5 pp decline in the share of middle-income peers and a 4.5 pp decline in the share of lower-income peers, for the average lower-income student.

a one-standard deviation (2 percentage-points) increase in upper-income peers increases college enrollment by about 0.11 percentage points (0.3%). I find evidence that enrollment rises in both two-year and four-year colleges.[48]

In a specification that includes only school and cohort fixed effects (i.e., excluding school trends) (Table 4, Model (1)), the estimated effect is slightly larger, but consistent: a one-SD (2 pp) increase in the share of upper-income peers is associated with a 0.18 percentage point higher likelihood of college enrollment.[49] I find no strong evidence of an impact on college selectivity (the estimated effect is consistently positive but not statistically significant) (Table 4). The estimated effect on college graduation is also consistently positive (roughly half the size of the effect on college enrollment) but not statistically significant in the preferred specification using school-specific linear trends (Table 5).

The impact of upper-income peers on lower-income students' college enrollment may interact with other constraints on accessing higher-quality colleges, including geographic proximity to colleges which Acton, Cortes, Miller, and Morales (2025) find impacts students' college enrollment. I find some evidence that distance to a four-year college might moderate the effect of upper-income peers: among students induced to enroll in college by exposure to a higher share of upper-income peers, those attending high schools closer to a four-year institution are more likely to enroll in selective colleges and to graduate (Table A10).[50]

I also find evidence of positive longer-run effects on average wages and income percentile rank.[51] The average wage of lower-income students in 2023 (ages 23-28) rises by \$87 (0.4%) for a one-SD increase (2 pp) in the share of upper-income peers ($p = 0.01$) (Table 5, Model (3)). Estimates are similar, though slightly smaller, in specifications without school trends (Table 5, Model (1)).[52]

48. The two-year and four-year college outcomes are based on "any" two-year and four-year college enrollment and so there could be overlap. As shown in Table A9, the positive impact on college enrollment seems to be driven by having more students enrolling in two-year and then enrolling in four-year colleges.

49. The standard deviation in the share of upper-income peers between cohorts without including school trends is slightly higher at 3 pp, but for consistency, I am using the standard deviation including school trends of 2 percentage points.

50. The minimum driving distance from high school calculation is based on Acton, Cortes, Miller, and Morales (2025) for high schools in 2023, which they generously shared. The distance from college varies very little year to year and so the minimum distance in 2023 should be representative of students' available options in earlier years as well (Miller, 2025). I do not find evidence of an interaction with any minimum distance to college (regardless of type).

51. Income percentile rank is based on ranking the full sample of students in a given cohort based on their wage in 2023. Students who do not have wage data are assumed to earn 0 that year.

52. The estimate on wages is based on imputing students who are missing unemployment insurance data with 0. If instead I only run the regression on students with employment data (61% of lower-income students) the wage gain is \$109 for one-SD (2 pp) increase in the share of upper-income students—a 0.3% increase from the control mean for students with employment data of \$33,743. The share of upper-income students does not seem to impact students' likelihood of being in the unemployment

Another policy-relevant way of scaling the estimates is using the variation in the share of upper-income students between schools in a district and between districts. A standard deviation difference in the share of upper-income peers is 12 pp between schools in a district and 20 pp between districts. Under the strong assumption that the within-school cohort estimates identified in this paper (from much smaller composition shocks) extrapolate linearly to these larger differences, the estimates imply attending a school in the same district with a one-SD higher share of upper-income peers (12 pp) would increase increase college enrollment by 0.7 percentage points (1.9%) and wages by $521 (2.5%). Alternatively, attending a district with a one-SD higher upper-income share (20 pp) would increase college enrollment by 1.2 percentage points (3%) and wages by $868 (4.2%). I discuss the scalability and external validity of these estimates in Section 5.5, as well as present some bounding exercises.

The estimated positive effects on college enrollment and wages appear to be driven by exposure to upper-income peers in students' *own* cohort, rather than by the composition of nearby cohorts. When I assign students the upper-income share of adjacent cohorts instead (Tables 4 and 5, Models (5) and (6)), the estimated effects on both college enrollment and wages are small and statistically insignificant.[53] The falsification test suggests the results reflect students' own cohort composition and provides no evidence of spillovers from adjacent cohorts.

Figure 7 visualizes the estimated relationship between cohort income composition and lower-income students' outcomes. In each panel, I plot residualized outcomes against residualized cohort shares of upper-income students. Residuals are constructed from regressions that include school and cohort fixed effects and control for student demographics and grade 8 test scores, corresponding to the baseline specification (Models (1) in Tables 4 and 5). In Panels (b) and (f), residuals additionally include school-specific linear trends, matching the preferred specification (Model (3) in Tables 4 and 5). Panels (c), (d), (g), and (h) implement the placebo tests by assigning students the upper-income share of adjacent cohorts (lead/lag), with residuals constructed including school trends (Model (5) and (6) in Tables 4 and 5).

Two patterns emerge. First, there is a clear positive, seemingly linear—relationship, between

insurance data.

53. The only exception is four-year college enrollment for the $t-1$ placebo, but it is not mirrored in the wage results and may be spurious.

within-school deviations in the share of upper-income peers and lower-income students' college enrollment and wages. Second, the relationship between peer income and outcomes disappears in the placebo panels when students are assigned the composition of older or younger cohorts, consistent with the interpretation that the effects are driven by students' own cohort rather than spillovers across cohorts.[54]

The positive impact of exposure to upper-income students on wages likely does not operate exclusively through increased college enrollment. Mountjoy (2022) using distance to two-year colleges finds that free or reduced lunch status students induced to enroll in a two-year colleges earn \$1,927 more in quarterly earnings.[55] Mountjoy (2022) reported wage gain suggests that an increase in two-year college enrollment of about 0.11 percentage points, for one-SD (2 pp) increase in the share of upper-income students, would increase yearly earning by \$8, almost one-tenth the estimated wage gain I observe in the paper. As shown in Table A9 most of the increase in college enrollment is driven by an increase in students enrolling in two-year and then four-year college. As an alternative benchmark, Zimmerman (2014) finds using university admissions cutoffs that the marginal free or reduced lunch status students' admission to a public university in Florida increases their four-year college enrollment by 14 percentage points and quarterly wages by \$950.[56] The increase documented by Zimmerman (2014) for four-year college enrollment suggests that an increase in 4-year college enrollment of about 0.10 percentage points for one-SD (2 pp) would increase yearly earning by \$27, which is still about one-third of the gain I observe in the paper. Together, both estimates suggest the impact on wages likely does not operate exclusively through changes in college enrollment.[57]

I find suggestive evidence that lower-income students enroll in colleges with higher average peer income: a one-SD (2 pp) increase in the grade 9 cohort share of upper-income peers is associated with enrolling in a college where average peer income (reported FAD) is about \$1,890 higher, though the estimate is not statistically significant at conventional levels ($p = 0.08$) (Table A12, Model (1)).

54. I formally test for nonlinearity by including the square of the share of upper-income peers. There is some evidence the impact on two-year college enrollment may be decreasing in the share of upper-income peers (Table A11). There is no strong evidence of nonlinearity in other outcomes.

55. I adjust the reported MTE quarterly earnings gain for disadvantaged students reported in 2010 dollars for inflation in 2023 US Dollars. The average is based on individuals 28–30.

56. Adjusted for inflation, assuming the \$737 wage gain reported in Zimmerman (2014) is based on 2014.

57. Another potential explanation is that students induced to enroll in college because of the share of upper-income peers have much higher returns than those documented by Zimmerman (2014) and Mountjoy (2022).

I also find positive but imprecise estimates for graduating with higher-paying majors (Table A12, Model (3)). These results are conditional on college enrollment (and on graduation for the major outcome) and should be interpreted as suggestive rather than conclusive.[58]

The relationship between the share of upper-income peers and long-term outcomes may operate through high school test scores and/or changes in course taking patterns. I find that a one-standard deviation increase in the share of upper-income peers increases lower-income students' average reading test score by 0.004 standard deviations (1.6%) (Table 6, Model (3)).[59] We also find evidence that lower-income students are more likely to take advanced courses and AP courses when in a cohort with a higher share of upper-income students (Table 6, Model (3)). Though the impact on the number of advanced courses is sensitive to the exclusion of school-specific time trends.

That said, I do find evidence of an increase in the level of within-school sorting by income by 0.28 percentage points for one-SD (2 pp) increase in the share of upper-income students. A 10-percentage point increase in the share of upper-income students in a school-cohort raises the share of upper-income classmates for lower-income students by 5.9 percentage points on average (Table 6, Model (3)), suggesting that lower-income students' direct classroom exposure to upper-income peers is much smaller, almost half of what would have been expected in the absence of sorting by income within schools, across classrooms.

The impact of higher-income peers could also operate through noncognitive skills. Ideally I would capture upper-income students' peer effects on Social and Emotional Development (SED) outcomes directly. Jackson, Porter, Easton, Blanchard and Kiguel, (2020) find that schools with a positive value add on students' SED, increase students' high school completion and college enrollment. The mediating effect of SED seems to be particularly important for disadvantaged students (Jackson, Kiguel, Porter and Easton, 2024). In the absence of a direct measure of peer income's effect on SED, I can capture the impact of having a higher share of upper-income peers on lower-income students' behavioral outcomes like attendance and suspension. I find that a one-SD increase in the share of upper-income students decreases students' out-of-school and in-school suspension by about 0.10 and 0.14 percentage points, respectively (Table 7, Model (3)). I also find evidence of a

58. College major expected return is based on linking students' major fields to ACS field of degree expected income in 2022 based on the income earned by various age groups.

59. The impact on test scores may be driven by a change in the composition of students taking the test. I do not find evidence that the change in peer income composition impacted the share of students missing reading test scores in high school.

0.022 percentage points decrease in the share of days absent for a one-SD increase in the share of upper-income students. I find no evidence of a change in the likelihood a student is retained the following year (coefficient is small and insignificant).

5.4 Mechanisms: Is it really about income?

The positive impact of having a higher share of upper-income peers on lower-income students may be mediated by other things correlated with peer income like achievement and race. Exposure to upper-income peers is highly correlated with exposure to higher-performing peers (correlation 0.75). As such, we may be capturing the impact of exposure to high-achieving students, regardless of family income. Similarly, the share of upper-income peers is highly correlated with the share of White peers (correlation 0.77).

To disentangle the role of peer achievement, I first run a regression that controls for peer achievement. In line with Section 4.2, high-achieving students are defined as those who performed in the top quarter of their grade 8 reading test score so that it is equivalent to the share of upper-income students. If peer income impacts college enrollment only through peer test scores, the coefficient on the proportion of upper-income students would approach zero when I control for peer test scores.

I find that the impact of exposure to upper-income peers on college enrollment remains positive and consistent with previous estimates when controlling for peer test score (Tables 5 and 4, Models (2) and (4))—if anything, it appears to be slightly larger. The consistent, slightly larger, estimates on the proportion of upper-income students suggest that there is something about being exposed to upper-income peers that impacts student college enrollment and wages through mechanisms other than peer achievement (orthogonal to peer test scores). The patterns are consistent independent of how higher-achieving students are defined: using math (instead of reading) grade 8 test scores, average math and reading, grade 3 and/or average grades 3 to 8 test scores as well as a continuous measure of peer achievement as shown in Table A13.

To better understand how peer income may interact with achievement, I split peers into four categories based on income and achievement. The positive impact of exposure to upper-income peers appears to be driven by exposure to lower-achieving upper-income students. One-SD (2 pp) increase in the share of upper-income, lower-achieving peers increases lower-income students' college enrollment by 0.4 percentage point ($p < 0.001$) and wages by \$96 ($p = 0.016$) (Table 8).[60] In contrast, one-SD increase in the share of upper-income higher-achieving peers slightly decreases lower-income students' college enrollment (-0.11 pp, $p = 0.06$) and has no impact on lower-income students' wages (\$31, $p = 0.48$) (Table 8).

The pattern in Table 8 is consistent with lower-achieving peers being more beneficial for college enrollment and wages (regardless of peer income) for the average lower-income student, and upper-income peers being more beneficial (compared to peers of similar test scores) for the average lower-income student, both relative to the average middle-income (sometimes on FRPL) peer.[61]

To assess whether similarity in achievement matters for the impact on college enrollment and wage, I re-estimate the models from Table 8, this time categorizing upper-income peers by their test score proximity to lower-income students. Specifically, I identify three subgroups of upper-income peers: (1) those scoring in the same quartile of grade 8 reading scores as the focal student, (2) those scoring one quartile above or below, and (3) those scoring two quartiles apart. Figure 8 summarizes the estimated coefficients, revealing that the positive impacts on both college enrollment and wages are concentrated among upper-income peers with similar test scores.

Students who are closer in achievement may be more likely to share courses and classrooms, and thus more likely to influence one another. I find that classroom exposure rises only slightly more for a one-SD (2 pp) increase in the share of upper-income, lower-achieving peers (1.2 pp) than for upper-income, higher-achieving peers (1.1 pp) (Table A14). The larger difference appears in advanced course-taking: a one-SD (2 pp) increase in exposure to upper-income, lower-achieving peers raises lower-income students' enrolled number of advanced courses by about 1%, whereas exposure to upper-income, higher-achieving peers has a negative but statistically insignificant effect on lower-income students' advanced course taking (Table A14).

Similar to achievement proximity, racial/ethnic proximity may increase interaction with, and po-

60. Lower-achieving students are defined as those who scored below the top quartile in their grade 8 reading test.

61. The estimates are consistent with the notion that peer achievement might have a negative impact on lower-income students. In Tables A4 and A5 I present a horse-race between the impact of the proportion upper-income and the proportion high-achieving students in a cohort. The impact of exposure to high-achieving students appears to be negative. This is consistent with a number of prior work (e.g., Abdulkadiroğlu, Angrist, and Pathak (2014), Feld and Zölitz (2017), Barrow, Sartain, and De la Torre (2020), and more recently Cattan, Salvanes and Tominey (2025)) who find that having higher-achieving peers have little or negative impact on students. The negative spillover from higher-achieving peers is unlikely to be driven by Texas's Top Ten Percent policy as there are similar negative effects of high-achieving peers in settings without that policy (e.g., Cattan, Salvanes, and Tominey, 2025). Moreover, Black, Denning, and Rothstein (2023) find no evidence that the Top Ten Percent rule reduces college enrollment or graduation among students who do not qualify for the top decile.

tential benefits from, upper-income peers. To examine if the impact is stronger within racial/ethnic groups, I re-estimate Equation (1) separating exposure to upper-income peers of the *same* race/ethnicity versus a *different* race/ethnicity, and I estimate these models separately for Black, Hispanic, and White lower-income students (Tables A15 and A16).

The wage results are substantially stronger for White students exposed to upper-income White peers: one-standard deviation (2 pp) increase in the share of upper-income White students raises wages by $219 relative to a much smaller and insignificant impact when exposed to higher-income peers of a different race/ethnicity. For college-going, the four-year enrollment effect is notably larger for Black students exposed to upper-income Black peers: one-standard deviation increase (2 pp) raises four-year college enrollment by 1 percentage points, relative to a much smaller impact when exposed to higher-income peers of a different race/ethnicity (0.2 percentage points). One exception is wages for Black students, where the estimates suggest larger gains from exposure to upper-income peers of a different race/ethnicity.

Consistent with the direct classroom exposure mechanism, lower-income students are slightly more likely to share a classroom with an upper-income student of the same race/ethnicity (Table A17). Overall, the larger effects for exposure to upper-income lower-achieving peers and to upper-income same-race peers suggest that proximity along salient dimensions may strengthen the upper-income peer impacts on lower-income students' college enrollment and earnings outcomes.

Other potential mechanisms besides achievement that are correlated with peer income such as peer behavioral changes (disciplinary record) and changes in racial composition may be driving the positive impact on lower-income students, rather than peer income. Increasing the share of upper-income students may decrease the share of disruptive peers whom we know from prior literature may have a negative impact on student outcomes (Carrell, Hoekstra and Kuka, 2018). To assess whether these factors can account for the results, I re-estimate Equation (1) replacing the share of upper-income students with (i) the share of peers with any grade-8 disciplinary record (any in-school or out-of-school suspension) and (ii) the share of White peers (Tables A18 and Table A19, Models (1) and (3)). I find no evidence that either peer disciplinary composition or the share of White students affects college enrollment or wages. This suggests that the estimated effects are not driven by correlated changes in disciplinary, racial, or achievement-related peer composition, but rather capture something else about exposure to upper-income peers.[62]

5.5 Addressing Economic Segregation and the Role of Access to School Resources

The positive impact of temporary small changes within a school in the share of upper-income students on lower-income students' outcomes suggests that the gains to improving cross-income exposure likely do not operate exclusively through changes in access to school resources. I examine whether lower-income students' access to experienced teachers and per pupil spending shifts with peer-income composition and find no detectable effects on teacher experience or per-pupil spending; the coefficients are small and statistically indistinguishable from zero (Table 9). I also find no evidence of a change in the number of advanced courses offered. That said, upper-income peers may affect access to unobserved resources (e.g., parent volunteering for school events or elective courses).

Because upper-income peers have positive effects on lower-income students—and these effects do not appear to operate solely through changes in observable school resources—closing the exposure gap to upper-income peers documented in Section 4 may be particularly important. Doing so necessarily implies that upper-income students are exposed to more lower-income peers. I find evidence suggesting that increasing cross-income exposure (i.e., swapping upper- and lower-income students) could be win–win or roughly neutral for early-adulthood wages. A 2 p.p. (one SD) increase in the share of upper-income students, holding constant the share of middle-income students (i.e., swapping upper-income for lower-income students), raises lower-income students' early-adulthood wages by 0.54% ($p = 0.002$; Table A20).[63] Conversely, a 2 p.p. increase in the share of lower-income students (holding the middle-income share fixed) has no statistically detectable effect on upper-income students' wages: the point estimate is -0.18% ($p = 0.30$; Table 10). While there is evidence of a small decline in upper-income students' graduation (a 0.3% decrease for a 2 p.p. increase in the

62. The estimated impact of upper-income peers is also consistent when controlling for the share of White students and the share of students with disciplinary records(Tables A18 and Table A19, Models (2) and (3)).

63. Table A20 isolates relative substitution of lower-income peers for upper- or middle-income peers: it compares replacing a lower-income peer with an upper-income peer versus replacing a lower-income peer with a middle-income peer. The estimates indicate that it is substituting lower-income for upper-income (not middle-income) peers that drives the increases in college enrollment and wages, suggesting the positive impact of upper-income peers is not simply about having fewer lower-income peers.

lower-income share), the decrease in graduation rate does not seem to translate into lower early-adulthood wages. Given the standard errors, I can rule out wage declines larger than 0.55% for a 2 p.p. increase in the share of lower-income students. That said, note that the decline in graduation rate may manifest in wages later on in life. Under a 10–percentage-point swap counterfactual (which would bring the average lower-income student up to the income integration benchmark), the estimated slope implies a 6% reduction in the upper–lower income gap in quarterly wages in early adulthood.

Considering policy implications, an important limitation of these findings is that a large-scale reduction in between-school income sorting could induce changes in school resources and family behavior that are not captured by my within school estimates. The identifying variation in this paper holds fixed the school a student attends (and therefore largely holds fixed school-level inputs), so the estimates should be interpreted as the effect of changing peer-income composition conditional on the existing resource environment. Integration policies that reassign students across schools may change that environment through shifts in staffing, course offerings, discretionary fundraising, or political support in ways that could either amplify or offset the peer-composition effect estimated here. The estimates in this paper do not capture (i) the effects on students who would need to change schools under a desegregation policy, (ii) any general-equilibrium responses (e.g., residential sorting in response and overall public school enrollment), or (iii) resource reallocation changes across schools that could accompany a large-scale reduction in between-school income segregation. For these reasons, the estimates should be interpreted as local average effects of modest, short-run changes in cohort composition rather than a complete forecast of the impacts of eliminating segregation across schools.

Temporary shocks used here may plausibly provide a lower bound on the gains from sustained changes in exposure if longer-run changes translate into durable improvements in access to school resources, course offerings, and/or information about postsecondary options. The lower-bound assumption, however, relies on the marginal effect of exposure being approximately stable as exposure changes—an assumption that would fail if effects are nonlinear. For example, the impact of having more upper-income peers may exhibit diminishing returns if the increase in the share of upper-income peers matters more when there are very few upper-income peers than when there are many. Using the data available, I do not find strong evidence for non-linearity by baseline school composition when interacting the share of upper-income peers by the schools' average share of upper-income students as shown in Table A21; the coefficients are generally negative on the interaction with the school's average share of upper-income students but insignificant. I also do not find strong evidence of nonlinearity across the observed range of cohort shocks (approximately −5 to +5 percentage points), as discussed in Section 5.5. That said, these tests cover only modest changes in exposure, and I may be underpowered to detect heterogeneity by schools' income composition.

To provide a sense of the magnitude of effects under larger and more persistent differences in exposure, I also estimate the impact of having a higher share of upper-income peers using specifications that exploit between-school variation in the share of upper-income peers. Between school variation is more vulnerable to selection: lower-income students attending schools with higher upper-income shares may differ in unobserved ways from those attending more economically disadvantaged schools. I therefore treat these estimates as suggestive bounds rather than causal policy forecasts. Table A22, Model (1) controls for student test scores and demographic characteristics and includes cohort fixed effects. Model (2) adds middle-school-by-cohort fixed effects, identifying the relationship from differences in high school upper-income share among students who attended the same middle-school cohort. Model (3) adds district fixed effects, identifying from differences among students within the same district who are in different cohorts and/or attend different high schools with different upper-income shares.[64] The estimates based on the larger between-school variation in the share of upper-income peers are very similar to those from the main within-school specification (Table 5).

Conditional on the assumptions required to extrapolate from local, within-school cohort shocks–and using the range of estimates based on the (potentially more biased) between-school variation in Table A22—I construct bounds on the potential wage gains for lower-income students from reducing economic segregation across schools and districts. The estimates imply that eliminating within-district segregation across schools would raise lower-income students' wages by about 0.5% to 0.7%, while eliminating both within-district and between-district segregation would raise average

64. A standard deviation in the residual variation in the proportion of upper-income students ranges from 15 to 8 percentage points in Table A22, Models (1)–(3), compared to only 2 percentage points in our main analysis using only variation within school including school trends.

lower-income students' wages by about 2.2% to 2.8%.

Finally, I find that the estimated impact on lower-income students' income is similar to that from Chetty et al. (2022), providing an external benchmark for the plausibility of the scale of the effects, though they use a different identification strategy (movers design) and focus on neighborhood rather than schools. Using movers design, Chetty et al. (2022) find that moving to a neighborhood with 1 unit higher economic connectedness increases relative mean income rank in adulthood by 9.8 percentile points. The slope of the relationship suggests that a 10 percentage-point difference in a schools' proportion upper-income peers (equivalent to a 0.16 units difference in economic connectedness) would increase mean income rank in adulthood by 1.6 percentile points.[65] The estimated increase in mean income rank is based on the slope of the relationship between years of exposure and income percentile—it assumes 20 years of exposure (from birth). Assuming the relationship between years of exposure and income percentile is linear up to 20 years of age, then exposure in high school years (four years) would yield a 0.3 percentile rank increase in mean income rank in adulthood which is similar to the increase in mean income rank in this paper of 0.36 percentile ranks.[66]

Though the estimated impact is similar to that in Chetty et al. (2022), evidence from desegregation interventions that explicitly change school assignment (e.g., court-ordered bussing) generally finds much larger positive impacts for historically disadvantaged students (Setren, 2025a; Bergman, 2018). Prior work also finds limited evidence that such large integration policies induce substantial exit among White families and finds little evidence of negative impacts on White students (Setren, 2025b; Tuttle, 2019; Angrist and Lang, 2004), which is consistent with the null effects I estimate for higher-income students' wages and suggests these nulls may be informative for broader integration policies as well.

When considering policy implications, it is important to account not only for changes in resources and potential political pushback, but also for endogenous sorting in response to shifts in student

65. The conversion from economic connectedness units to school proportion of upper-income students is based on the slope of the relationship between a schools' economic connectedness as captured by Chetty et al. (2022) and average lower-income students' proportion of upper-income classmates as captured in Section 4.3.

66. To compare the estimates of the impact of economic connectedness to the estimates of the impact of exposure to upper-income peers in this paper, we need to assume that for a lower-income student, the effect of a 10-percentage-point change in the proportion of upper-income peers in the same school is equivalent to moving to a *neighborhood* with a 10-percentage-point difference in the proportion of upper-income peers. The 9.8-percentile points estimate can serve as a rough benchmark, but ultimately it is capturing differences between neighborhoods that could include differences in school quality and neighborhood institutions. In comparison, the estimates in this paper only capture differences between peers in the same school.

composition. Carrell et al. (2013) show that increasing heterogeneity in peer achievement can generate negative effects when students re-sort into more homogeneous groups within the assigned setting.[67] Consistent with the endogenous sorting concern, I find evidence of increased within-school sorting by income across classrooms even for the small composition shocks used in this paper: a one-standard-deviation increase in the cohort share of upper-income students (2 pp) increases the within-school classroom sorting measure by 0.27 percentage points or 3% (Table 6).[68] As such, policies designed to increase lower-income students' exposure to upper-income peers may need to consider course-taking and classroom assignment patterns—especially when income differences align with race or prior achievement. As shown earlier, students are slightly more likely to share a classroom with an upper-income student of a similar race and/or achievement (Tables A14 and A17).

5.6 Robustness Checks: Measurement Error and Endogenous Sorting

Deviations in the share of upper-income students may reflect measurement error in FRPL status rather than true changes in cohort composition—in other words, it could be that lower-income students' peer income composition did not change it just happens to be that one student who should have been assigned one year on free/reduced lunch was not assigned to free/reduced lunch due to administrative or other errors. To assess this, I test whether these deviations predict alternative income measures. A one-SD (2 pp) increase in the share of upper-income (never-FRPL) peers is associated with about $2,000 higher average cohort parental income as reported on financial aid applications and 2 pp fewer cohort-average years on FRPL (Table A23). These patterns suggest that within-school cohort deviations in upper-income share capture meaningful differences in peers' parental income, not just FRPL misclassification that may be correlated with students' own outcome.

We can combine the estimated relationship between the cohort share of upper-income (never on FRPL) peers, and the average parental income reported in FAD to express the peer-income composition effects in "dollars of peer parental income". Doing so requires two strong assumptions:

67. Setren (2025a) also finds evidence of tracking when Black students were bussed to majority White schools.

68. Classroom sorting is calculated using the variance ratio which captures the difference between the average lower-income students share of upper-income classmates and the average upper-income students' share of upper-income classmates, in the same school.

(1) that the share of upper-income (never-FRPL) peers only affects outcomes through changes in the (true) income distribution of peers; and (2) the average FAD reported income represents the true change in the average peer income. Assumption (2) is unlikely to hold because, as shown in Table A23, lower-income students are more likely to submit FAD applications in cohorts with a higher share of never-FRPL peers, which could mechanically reduce the observed average parental income among FAD filers without changing the true distribution of parental income. With these caveats, the implied scaling suggests that exposure to peers with parental income \$10,000 higher increases college enrollment by 0.5 pp (0.8%) and yearly earning by \$387 (1.9%) at 23–28.

Another income measurement concern is that students' likelihood of being on free/reduced lunch in high school may be correlated with the share of upper-income students in the cohort. For example, lower-income students may be less likely to enroll in free/reduced lunch when in a cohort with a higher share of upper-income students. If so, then we may be capturing differences in selection into free/reduced lunch that are correlated with the share of upper-income students. To address this concern, I re-estimate the regression using a definition of lower- and upper-income students based on their free/reduced-price lunch status in the years prior to high school.[69] When only using prior to high school years to define income, I find a similarly positive impact on lower-income students' college enrollment and wages as shown in Table A24. The impact on college enrollment remains consistent in size and precision, and and the impact on wages is positive, but smaller and less precise (\$44 (p =0.06)increase for a one-SD (2 pp) change in the share of upper-income students). By only using prior to high school years we are potentially missing important information on students' degree of economic disadvantage (introducing attenuation bias), but it is less susceptible to composition-induced changes in FRPL take-up during high school.

The second key threat to identification is endogenous sorting of lower-income students across schools (or out of the public-school sample) in response to cohort-level deviations in income composition. For instance, more motivated lower-income families might be more likely to remain in (or move into) cohorts with a higher share of upper-income peers. If so, lower-income students' potential outcomes could be correlated with the cohort share of upper-income students, biasing the

69. Twenty-six percent of students who consistently receive free/reduced-price lunch prior to high school are no longer recorded as on free/reduced lunch in one or more high school years. The switch to no longer being on free/reduced lunch in high school may reflect real improvements in household income, as parents may earn more as students grow older, or changes in the students' likelihood of applying for free/reduced lunch.

main estimates. Two pieces of evidence suggest this is unlikely to be driving the results: First, if sorting were occurring, we would expect lower-income students in cohorts with more upper-income peers to look different on observables. As discussed earlier, lower-income students in cohorts with a higher upper-income share do not differ meaningfully in demographic characteristics or parental-reported income in FAD as shown in Table 3. Though there remain concerns lower-income students may differ on unobservable characteristics like parental involvement. Second, I can bound the estimates using information we have on students' mobility in 9th to 10th grade in response to peer composition. To assess whether lower-income students respond to cohort composition by changing schools, I estimate the relationship between upper-income peer share and the probability that a lower-income student leaves their initial school after ninth grade. The estimate is small and statistically insignificant: a one-SD (2 pp) increase in upper-income peers is associated with a 0.04 pp decrease in switching schools ($p = 0.416$), suggesting little evidence of differential school switching in response to cohort income composition (Table A25).

That said, I do find evidence of differential sample attrition: a one-SD 2 pp higher upper-income peer share is associated with a 0.08 pp lower probability that a lower-income student leaves the sample (exits to private school, moves out-of-school or drops out) ($p = 0.005$) as shown in Table A25. The decrease in attrition implies that lower-income students in cohorts with higher-than-expected share of upper-income peers are slightly more likely to be retained in the Texas public school system. To assess whether differential attrition could mechanically generate our main estimate, we can conduct a simple back-of-the-envelope selection calculation. If the true causal effect on college enrollment were zero, then selection would need to account for the full +0.11pp increase in college enrollment for one-SD increase in the share of upper-income peers. This would require the marginal 0.08 percentage point of lower-income students who are "retained" in higher–upper-income cohorts (and would otherwise attrit) to have college-enrollment rates of more than 137 pp higher than those who would attrit—an impossibility since enrollment is bounded between 0 and 100 percent. On average, lower-income students who attrit enroll in college at a rate of 33 pp lower than lower-income students who remain in the sample.[70]

70. For the estimates of the effect of lower-income peers on higher-income students, a potential concern is differential attrition: if higher-income students who are most affected by exposure to more lower-income peers are more likely to leave the observed sample, estimates could be biased toward zero. However, I find the opposite pattern for within–public-school mobility; a one-SD (2 pp) increase in the cohort share of lower-income students reduces the probability that a higher-income student switches schools by 0.24 pp ($p = 0.003$), and there is no evidence of differential exit from Texas public schools (coefficient = 0.0018; p

6 Discussion and Conclusion

I present new evidence on the stark income-based disparities in students' classroom share of upper-income peers and evaluate its implications for lower-income students' college enrollment and wages. Using rich administrative data from Texas and a new way of measuring family income using repeated measures of free/reduced lunch, I show that lower-income students encounter far fewer upper-income classmates over their schooling years than their upper-income counterparts: cumulatively, the median lower-income student shares classrooms with fewer than 40 upper-income peers out of roughly 700 classmates between grades 5 and 12. Although the limited exposure to upper-income classmates is largely driven by residential and school-level segregation, classroom-level sorting, particularly in higher-income districts, accounts for a meaningful share and may be more amenable to policy intervention.

Using the residual from school trend variation in the share of upper income students across cohorts, I find that increased exposure to upper-income peers improves lower-income students' college enrollment and early adult wages. I show that these gains seem to be driven by exposure to upper-income, lower-achieving, students. These residual shocks in the share of upper-income peers are not correlated with changes in lower-income students' demographic characteristics and do not seem to change schools' (observable) resources (e.g. number of advanced courses offered and share of experienced teachers). The patterns indicate the benefits of cross-income exposure likely operate through channels other than the conventional school-level resource inputs and peer achievement. While evidence points toward social capital mechanisms, in this paper I do not directly measure information flows and peer networks. That said, I show that measures of cross-income friendships in Chetty et al. (2022) using Facebook data strongly align with administrative measures of the share of upper-income peers, and the estimated impact on lower-income students' income percentile.

The results underscore a policy-relevant dimension of educational inequality: cross-income exposure itself may matter, beyond the allocation of observable school inputs. One concern in integration debates is whether increasing exposure to lower-income peers harms upper-income students. I find

= 0.857). Higher-income switchers look very similar to non-switchers: switchers are only 0.04 percentage points less likely to enroll in college, on average, and earn about 8% less by 2023. Given the small switcher–non-switcher outcome gaps and the modest effect of cohort composition on switching, differential switching is unlikely to materially bias the estimated peer effects for higher-income students.

no evidence of declines in upper-income students' early-adulthood wages, although there is some evidence of a reduction in graduation. Overall, the wage results suggest that increasing cross-income exposure can generate mobility gains for lower-income students with limited short-run tradeoffs for upper-income peers.

These estimates are identified from relatively small, temporary cohort shocks. Extrapolating to large, sustained economic desegregation policies requires additional assumptions, as larger changes may affect resources and induce behavioral responses such as school switching or exit from the public system. That said, evidence from racial desegregation policies that explicitly changed school assignment (e.g., court-ordered busing) suggests that the gains from sustained integration can be larger for historically disadvantaged students (Setren, 2025A; Bergman, 2018), with limited evidence of negative effects on White students (Setren, 2025B; Tuttle, 2019; Angrist and Lang, 2004). Prior work also indicates that cross-class contact can increase prosocial attitudes among advantaged students, such as generosity and support for redistribution (Rao, 2019; Ethan, Jorge, and Cody, 2025), potential benefits not captured in this paper but consistent with broader social returns to integration.

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7 Main Tables and Figures

Table 1: Texas State Public School Student Demographics: Cohorts 2012-2014

Variable	Mean
Asian-American	.048
	(.213)
Black Students	.141
	(.348)
White Students	.320
	(.466)
Hispanic Students	.516
	(.500)
Enrolled in Title 1 School	.572
	(.495)
Free/Reduced Lunch Status	.608
	(.488)
Special Education	.094
	(.292)
Gifted Program	.106
	(.307)
English Language Learner	.262
	(.440)
Number of Students	1128554

This table summarizes demographic characteristics of the three cohorts of students I follow from grade 5 to expected grade 12. The demographic characteristics are based on student demographics (including free/reduced lunch status in fifth grade). The average is weighted by number of students across three cohorts. A cohort is defined by the year I observe them enrolled in grade 5 (2012, 2013 or 2014). The number in the brackets is the standard deviation.

Table 2: Variation of interest: Between Cohorts 2010 and 2019–Residual Change in Income and Achievement of 9th Graders

Statistic	High-Income	High-Achieve
Standard Deviation	.021	.039
1st Percentile	-.059	-.115
5th Percentile	-.029	-.058
10th Percentile	-.019	-.039
90th Percentile	.019	.04
95th Percentile	.03	.057
99th Percentile	.062	.104

This table summarizes the residual variation in the proportion of high-income students and the proportion of high-achieving students. High-achieving students are those who scored in the top quarter of their grade 8 reading test. The residual variation is based on Equation (1): it is the residual variation after including cohort and school-trend controls. It is based on the sample of low-income students, i.e., it is weighted by the number of low-income students enrolled, which is the population of interest here.

Table 3: Correlation Between Proportion Upper-Income Students in High School and Lower-Income Students' Demographic Characteristics

	(1)	(2)	(3)	(4)
Black Student	-0.0471	-0.0453	-0.00600	-0.00289
	(0.0156)	(0.0154)	(0.0134)	(0.0135)
Group Mean	0.150	0.150	0.150	0.150
N Clusters	2358	2358	2358	2358
White Student	0.0416	0.0405	0.0198	0.0202
	(0.0118)	(0.0119)	(0.0134)	(0.0134)
Group Mean	0.0900	0.0900	0.0900	0.0900
N Clusters	2358	2358	2358	2358
Hispanic Student	0.0110	0.00535	-0.00169	-0.00898
	(0.0200)	(0.0199)	(0.0186)	(0.0187)
Group Mean	0.680	0.680	0.680	0.680
N Clusters	2358	2358	2358	2358
Ever Immigrant	-0.0597	-0.0608	-0.0129	-0.00890
	(0.0159)	(0.0164)	(0.0114)	(0.0116)
Group Mean	0.100	0.100	0.100	0.100
N Clusters	2358	2358	2358	2358
Female Student	0.0279	0.0193	0.00189	-0.00360
	(0.0159)	(0.0159)	(0.0195)	(0.0196)
Group Mean	0.470	0.470	0.470	0.470
N Clusters	2358	2358	2358	2358
Income Reported on FAD	-8.773	9.883	5.782	14.56
	(23.39)	(24.03)	(23.69)	(23.73)
Group Mean	60.27	60.27	60.27	60.27
N Clusters	2153	2153	2153	2153
Predicted Wage	955.0	971.3	238.2	154.5
	(277.6)	(281.4)	(218.4)	(222.2)
Group Mean	20670.7	20670.7	20670.7	20670.7
N Clusters	2358	2358	2358	2358
Predicted College Enrollment	0.0179	0.0163	0.00361	0.00111
	(0.00550)	(0.00552)	(0.00476)	(0.00479)
Group Mean	0.370	0.370	0.370	0.370
N Clusters	2358	2358	2358	2358
School time trend			X	X
Proportion High-Achieving Peers		X		X

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1), but without any individual demographic and test score controls, Table A3 includes individual controls (leaving out the demographic we are testing balance on). The sample includes lower-income students only. Models (1) and (2) do not include school time trends. Ever immigrant is based on if a students was ever assigned an immigrant status in the Texas data. A student is labeled as immigrant in the data if they were not born in any state and have not attended any state for more than three academic years. Reported income is in 1000 dollars and is based on the subsample of lower income students who submit a financial aid application (34% of lower income students). The average reported income for lower-income students is much higher than the median because of a few outlier reported income; the median parent reported income for lower-income students is 20.2 thousand. The predicted wage and college outcomes are based on a linear regression on students demographic characteristics, the predicted wage coefficients are based on older cohorts (2010-2014) for whom we can observe wage outcomes at 23–28.

Table 4: Impact of Proportion Upper-Income Students on College Enrollment for Lower-Income Students

	Main effects				Falsification test	
					t+1	t-1
	(1)	(2)	(3)	(4)	(5)	(6)
Enrolled in College/University						
Proportion Upper-Income	0.0918	0.127	0.0555	0.0907	-0.00960	0.0241
	(0.0224)	(0.0226)	(0.0210)	(0.0210)	(0.0219)	(0.0206)
Group Mean	0.370					
Enrolled in a Public 2yr College						
Proportion Upper-Income	0.125	0.146	0.0460	0.0706	-0.0161	0.00725
	(0.0238)	(0.0239)	(0.0223)	(0.0228)	(0.0219)	(0.0219)
Group Mean	0.290					
Enrolled in a Public 4yr College						
Proportion Upper-Income	0.0340	0.0561	0.0491	0.0705	-0.00239	0.0425
	(0.0142)	(0.0139)	(0.0156)	(0.0155)	(0.0161)	(0.0162)
Group Mean	0.140					
Enrolled in a Selective University						
Proportion Upper-Income	0.00752	0.0131	0.00699	0.0125	0.00111	0.00423
	(0.00397)	(0.00412)	(0.00470)	(0.00482)	(0.00508)	(0.00534)
Group Mean	0.0200					
School time trend			X	X	X	X
Proportion High-Achieving Peers		X		X		
N	1501742	1501742	1501742	1501742	1324870	1348885
N Clusters	2358	2358	2358	2358	2235	2270

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). The sample includes lower-income students only. Models (1) and (2) do not include school time trends. College enrollment outcomes include cohorts 2010–2019. Models (2) and (4) include both the proportion of upper-income and higher-achieving peers in the cohort. Group Mean is the average for lower-income students in the sample. Models (5) and (6) are placebo tests using the proportion of upper-income students of cohort $t+1$ Model (5) or $t-1$ Model (6).

Table 5: Impact of Proportion Upper-Income Students on College Graduation and Wages for Lower-Income Students

	Main effects				Falsification test	
					t+1	t-1
	(1)	(2)	(3)	(4)	(5)	(6)
Graduated College/University						
Proportion Upper-Income	0.0541	0.0605	0.0249	0.0311	-0.00239	0.0115
	(0.0213)	(0.0211)	(0.0229)	(0.0229)	(0.0254)	(0.0297)
Group Mean	0.170					
Wage 2023						
Proportion Upper-Income	3190.6	3700.9	4337.8	5141.6	641.3	-2882.8
	(1409.3)	(1425.5)	(1670.4)	(1689.0)	(1797.5)	(1916.5)
Group Mean	20490.5					
Income Percentile 2023						
Proportion Upper-Income	1.879	3.020	3.560	4.412	0.576	-2.027
	(1.363)	(1.383)	(1.620)	(1.638)	(1.706)	(1.943)
Group Mean	46.75					
School time trend			X	X	X	X
Proportion High-Achieving Peers		X		X		
N	700245	700245	700245	700245	695382	560867
N Clusters	1992	1992	1992	1992	1967	1929

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). Models (1) and (2) do not include school time trends. Models (2) and (4) include both the proportion of upper-income and higher-achieving peers in the cohort. Group Mean is the average for lower-income students in the sample. College graduation and wages include cohorts 2010–2014. Models (5) and (6) are placebo tests using the proportion of upper-income students of cohort $t+1$ Model (5) or $t-1$ Model (6).

Table 6: Impact of Proportion Upper-Income Students on High School Course Taking and Academic Outcomes

	Model 1	Model 2	Model 3	Model 4
Share Upper-Income Classmates				
Proportion Upper-Income	0.701	0.710	0.592	0.600
	(0.00847)	(0.00824)	(0.00761)	(0.00749)
Group Mean	0.130			
N	1369434			
Classroom Sorting				
Proportion Upper-Income	0.157	0.158	0.135	0.136
	(0.0139)	(0.0140)	(0.0115)	(0.0114)
Group Mean	0.0900			
N	1300067			
Advanced Courses Taken				
Proportion Upper-Income	-0.421	-0.270	0.538	0.767
	(0.290)	(0.283)	(0.275)	(0.270)
Group Mean	2.270			
N	1501742			
AP Courses Taken				
Proportion Upper-Income	-0.644	-0.472	0.449	0.588
	(0.234)	(0.228)	(0.205)	(0.204)
Group Mean	1.280			
N	1501742			
Reading/English I Test Score				
Proportion Upper-Income	0.454	0.587	0.211	0.376
	(0.0830)	(0.0856)	(0.0782)	(0.0795)
Group Mean	-0.260			
N	1401373			
Missing Reading/English I Test Score				
Proportion Upper-Income	-0.0109	0.00306	0.0213	0.0232
	(0.0151)	(0.0139)	(0.0172)	(0.0176)
Group Mean	0.0700			
N	1501742			
School time trend			X	X
Proportion High-Achieving Peers		X		X

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). Group Mean is the average for lower-income students in the sample. Classroom sorting and classroom exposure regressions are based on the 2011–2019 cohorts, as classroom data was not available before 2011. Classroom sorting is based on capturing the difference in the average share of upper-income classmates between upper- and lower-income students attending the same school, as such it is missing if there are no upper-income students in the school. All other outcome regressions are based on the 2010–2019 cohorts. The test score outcome has fewer observations since 6% of students are missing their reading scores. Most students take the Reading/English I test in grade 9. For students who took the test in 2010 and 2011, their scores were based on the Reading TAKS test. For students who took the test after 2011, their scores were based on the English I STAAR test. All raw scores were standardized based on the distribution of students who took the test that year.

Table 7: Impact of Proportion Upper-Income Students on Behavioral Outcomes

	Model 1	Model 2	Model 3	Model 4
Grade Retention				
Proportion Upper-Income	0.00197	0.00698	-0.000186	-0.00218
	(0.0164)	(0.0159)	(0.0143)	(0.0144)
Group Mean	0.100			
N	1501742			
Proportion of Days Absent				
Proportion Upper-Income	-0.00971	-0.00313	-0.0111	-0.00977
	(0.00709)	(0.00718)	(0.00522)	(0.00520)
Group Mean	0.100			
N	1501742			
Any Out-of-School Suspension				
Proportion Upper-Income	-0.0304	-0.0308	-0.0479	-0.0534
	(0.0240)	(0.0240)	(0.0188)	(0.0187)
Group Mean	0.270			
N	1501742			
Any In-School Suspension				
Proportion Upper-Income	-0.0927	-0.0925	-0.0685	-0.0701
	(0.0356)	(0.0364)	(0.0249)	(0.0247)
Group Mean	0.420			
N	1501742			
School time trend			X	X
Proportion High-Achieving Peers		X		X

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). Group Mean is the average for lower-income students in the sample. A student in a cohort is considered retained if I observe them the following year in grade 9. Attendance and suspension outcomes are based on the share of member days and any suspensions in expected grades 9 to 12.

Table 8: Impact of Changes in Income and Achievement Composition on Lower-Income Students: 4 Groupings

	(1) Any College	(2) 2yr Public	(3) 4yr Public	(4) Selective	(5) Grad Any	(6) Wage 2023	(7) Income Percentile
Proportion Upper-Income and High-Achieve	-0.0542	-0.0226	-0.0237	-0.00775	0.0257	1554.6	0.0397
	(0.0285)	(0.0272)	(0.0208)	(0.00707)	(0.0311)	(2163.6)	(2.104)
Proportion Upper-Income and Lower-Achieve	0.181	0.143	0.124	0.0197	0.0302	4804.8	3.651
	(0.0255)	(0.0272)	(0.0180)	(0.00553)	(0.0274)	(1983.2)	(1.930)
Proportion Low-Income and High-Achieve	-0.0565	-0.0245	-0.0255	-0.0267	-0.0559	-7056.8	-8.215
	(0.0406)	(0.0431)	(0.0297)	(0.0102)	(0.0420)	(2307.9)	(2.240)
Proportion Low-Income and Lower-Achieve	0.0876	0.0806	0.0436	0.00776	0.0133	-1200.6	-2.304
	(0.0198)	(0.0201)	(0.0115)	(0.00322)	(0.0178)	(1127.7)	(1.133)
N Clusters	2358	2358	2358	2358	1992	1992	1992
N	1501742	1501742	1501742	1501742	700245	700245	700245

Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1) but instead of the main independent variable (proportion upper-income students), peers are split into four groups based on if they are upper-income and/or higher-achieving (test score in the top quarter of grade 8 reading test). Models (1)–(4) include cohorts 2010–2019 and Models (5)–(7) include cohorts 2010–2014.

Table 9: Impact of an Increase in the Proportion of Upper-Income Students on Access to School Resources

	(1) Teacher Experience	(2) Novice Teachers	(3) Spending	(4) Advanced Courses Offered	(5) AP Courses Offered
Proportion Upper-Income	0.433	0.00101	1804.2	0.529	0.163
	(0.363)	(0.0210)	(2226.9)	(0.575)	(0.350)
Group Mean	10.548	0.285	10605.050	10.895	6.122
N Clusters	2184	2184	1873	2358	2358
N	1219772	1219772	1161067	1501742	1501742

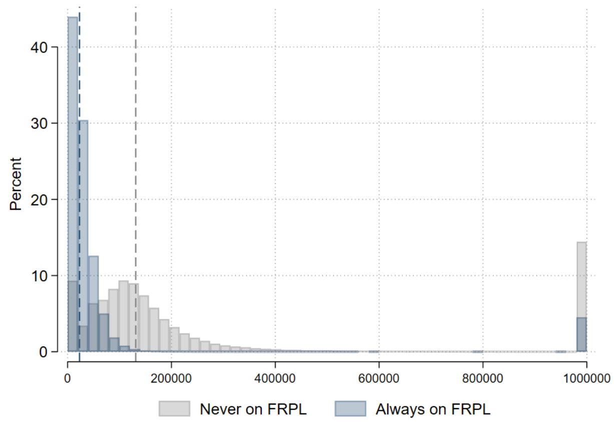
Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). Models (1)–(3) include cohorts 2012–2019 because teacher and school budget data linked to classrooms are only available starting in 2012. Spending is average school spending per pupil in a given year. Models (4)–(5) include cohorts 2010–2019 and capture the number of advanced and AP courses taken by a lower-income student, on average.

Table 10: Impact of Proportion Lower-Income Students, Holding Constant Mid-Income Students, on Upper-Income Students

	(1) Any College	(2) 2yr Public	(3) 4yr Public	(4) Selective	(5) Grad Any	(6) Wage 2023	(7) Income Percentile
Proportion Lower-Income	-0.0430	-0.0193	-0.0314	0.00610	-0.0808	-2970.7	-1.423
	(0.0286)	(0.0310)	(0.0278)	(0.0185)	(0.0414)	(3149.7)	(2.345)
Proportion Middle-Income	-0.0755	-0.0495	-0.0655	-0.00304	-0.102	-3564.9	-1.406
	(0.0231)	(0.0286)	(0.0228)	(0.0159)	(0.0333)	(2598.3)	(1.938)
Group Mean	0.680	0.490	0.420	0.160	0.470	33651.8	54.91
N Clusters	2129	2129	2129	2129	1818	1818	1818
N	1105943	1105943	1105943	1105943	549280	549280	549280

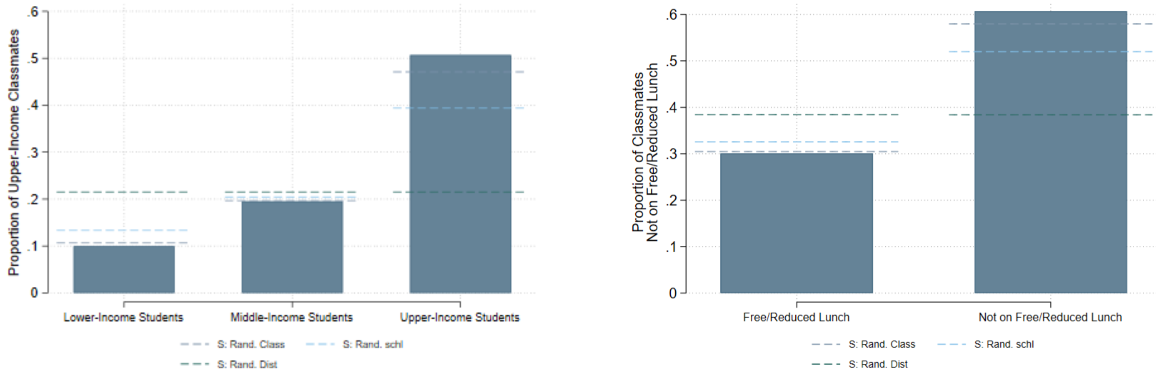
Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1) including a control for the share of middle income students in the cohort. The sample is limited to higher-income students. Models (1)–(4) include cohorts 2010–2019 and Models (5)–(7) include cohorts 2010–2014.

Figure 1: Distribution of Parental Income by Free/Reduced Lunch Status



Notes: The sample includes students students enrolled in grade 5 in Texas between 2012 and 2014 who applied for financial aid during expected college enrollment years (2017—2022). Parental Income is based on the adjusted parental gross contribution listed on a student's financial aid applications at a single point in time. The vertical lines are the median income for each group. Outlier students with parents who have listed an income above 999,999 are top-coded to 999,999 (in earlier years, parents could not report income above 999,999 due to limited character space). Financial aid data are not available for 66%, 59% and 49% of students always, sometimes, and never on free/reduced lunch, respectively.

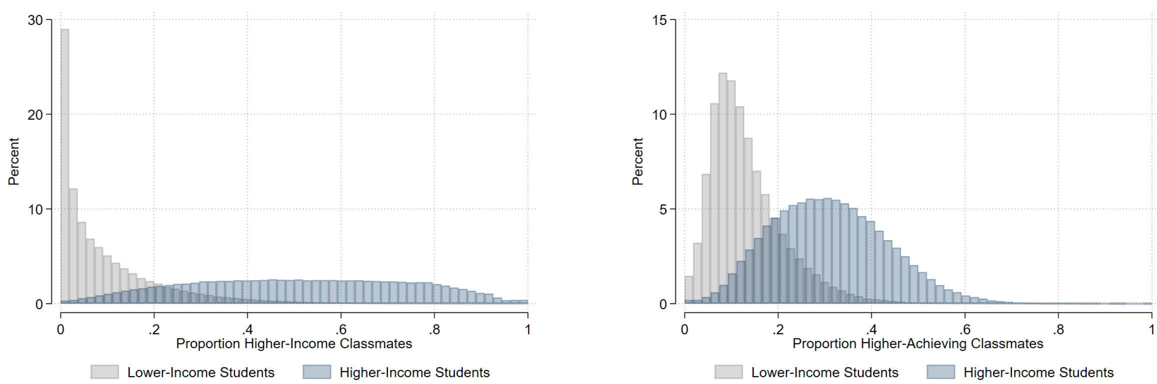
Figure 2: Exposure to Higher-Income Students: Observed Relative to Random Assignment



(a) Using Years on FRPL Status (b) Using Contemporaneous FRPL Status

Notes: The bars capture the cumulative share of classmates that are upper-income calculated as the total number of upper-income students a student is in a classroom with from grade 5 to expected grade 12 (excluding own status) divided by the total number of students in each classroom (excluding self). The average exposure is weighted by the number of students in each group. The dashed horizontal lines present the various benchmarks for integration. "S: Rand. Dist" lines capture the expected proportion of higher-income classmates had students been randomly assigned to districts: district integration benchmark. The "S: Rand. Schl" lines capture the expected proportion of higher-income classmates had students been randomly assigned to schools within a district (holding constant district composition): school integration benchmark. The "S: Rand. Class" lines capture the expected proportion of higher-income classmates had students been randomly assigned to classrooms within a school (holding constant school composition): classroom integration benchmark. In Panel (a) upper-income and lower-income students are defined as years on free/reduced lunch. In panel (b) not-on-free/reduced lunch students are students who in a given year are not on Free/Reduced Lunch status.

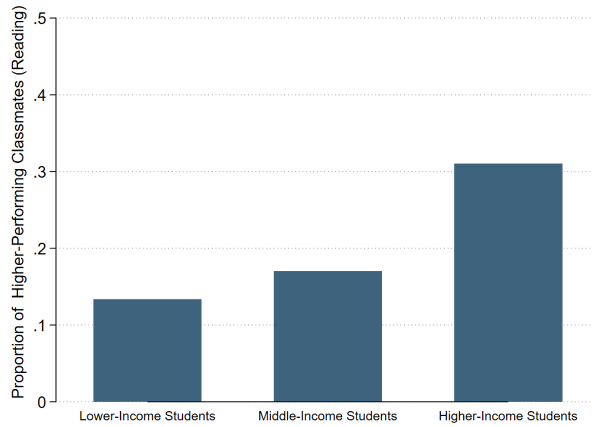
Figure 3: At the 25th Percentile, Lower-Income Students Have 1% Upper-Income and 7% Higher-Achieving Classmates



(a) Proportion of Upper-Income Classmates (b) Proportion of Higher-Achieving Classmates

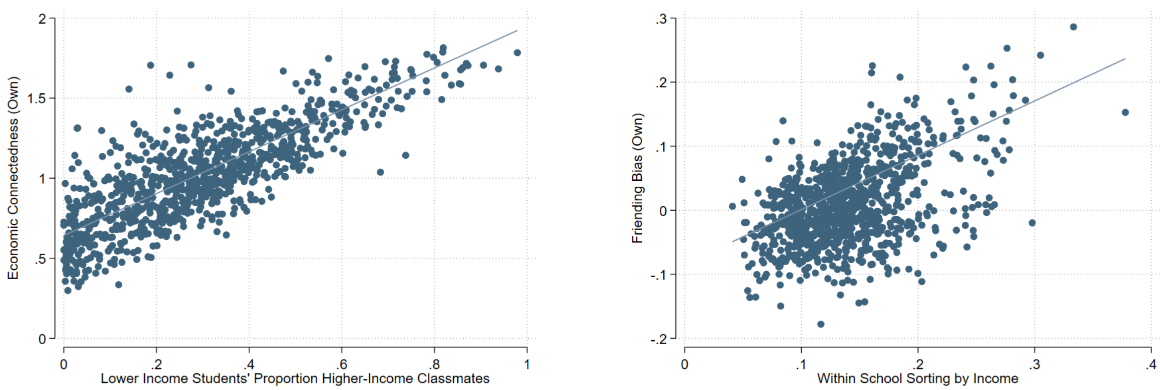
Notes: Every time a student is in a classroom with an upper-income/higher-achieving student, this is counted as one interaction. Histograms present the distribution of the proportion of classmates in cohorts 2012–2014 who are upper-income/higher-achieving between G5-12 for higher- and lower-income students. The average number of classroom interactions between grades five and twelve is approximately 2000.

Figure 4: The gap in Exposure to Higher-Performing (Top quarter) Students is Smaller than the Gap in Exposure to Upper-Income Students



Notes: This plot captures students' cumulative proportion of higher-performing classmates between grade 5 to expected grade 12. I define higher-performing students as students who performed in the top 24 percentiles of test score based on the grade 4 standardized reading test. The percentiles are based on the distribution of students who have a grade 4 test score. Approximately 5% of students are missing grade 4 reading test scores.

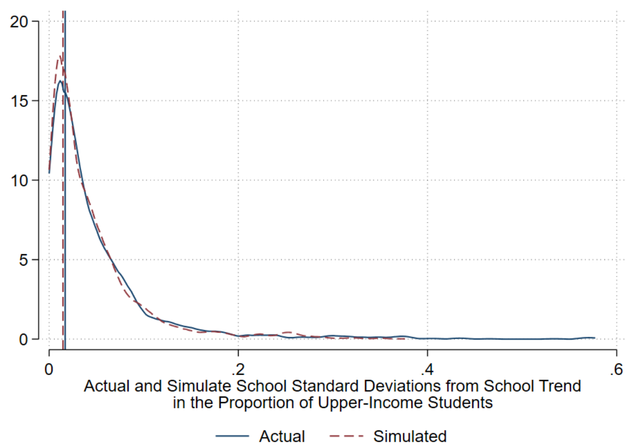
Figure 5: Strong Relationship between Likelihood of Befriending Higher-Income Students and Classroom Exposure



(a) Classroom Exposure and Economic Connectedness (b) Classroom Sorting by Income and Friending Bias

Notes: Each dot represents a school. Of the 1132 schools high schools linked the friendship formation high school measures in Chetty et al. (2022), 981 have data on economic connectedness. The fitted line weights schools by number of students enrolled. Panel (a) presents the correlation between high school measures of economic connectedness as captured by Chetty et al. (2022) and the proportion of higher-income students to whom lower-income students are exposed in each high school in 2012. Economic connectedness captures the likelihood of befriending a higher-income individual on Facebook. Panel (b) presents the correlation between high school measures of friending bias as captured by Chetty et al. (2022) and the difference in the proportion of higher-income students in higher- and lower-income students' classrooms, within a school (variance ratio). The variance ratio is based on 2012 student enrollment data. Friending bias captures the likelihood of befriending a higher-income individual on Facebook, conditional on the school proportion of higher-income students.

Figure 6: Simulated and Actual Distribution of within School Standard Deviations from School Trend

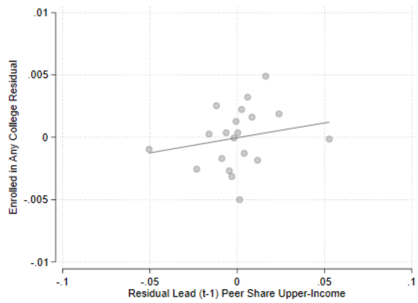


Notes: The figure captures the distribution of each school's standard deviation in the proportion of upper-income students from school trend (solid line) and each school's simulated average standard deviations in the proportion of upper-income students from school mean (dashed line). The vertical lines capture each distributions' median standard deviation in the proportion of upper-income students.

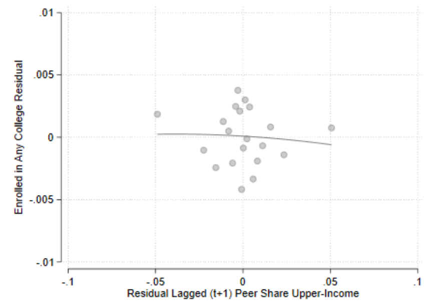
Figure 7: School-Cohort Variation in the Share of Upper-Income Peers and College Enrollment

(a) Residual College Enrollment

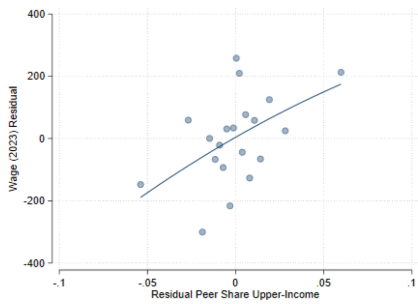
(b) Residual College Enrollment with School Trends



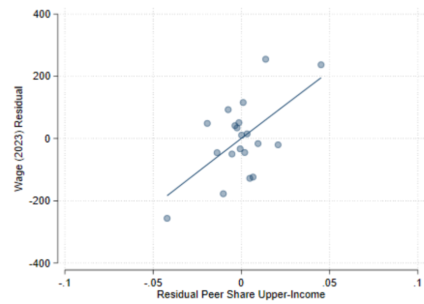
(c) Residual Lagged Cohort College Enrollment



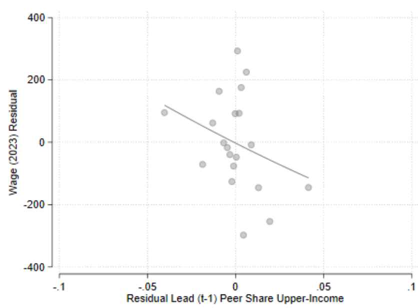
(d) Residual Lead Cohort College Enrollment



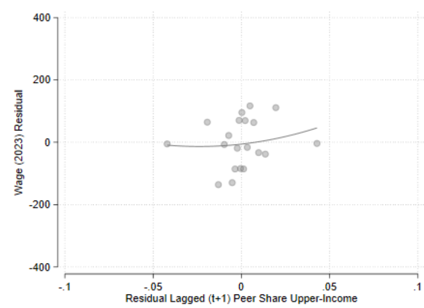
(e) Residual Wages



(f) Residual Wages with School Trends



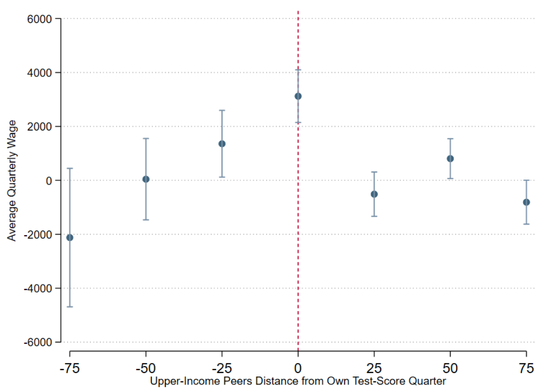
(g) Residual Lagged Cohort Wages



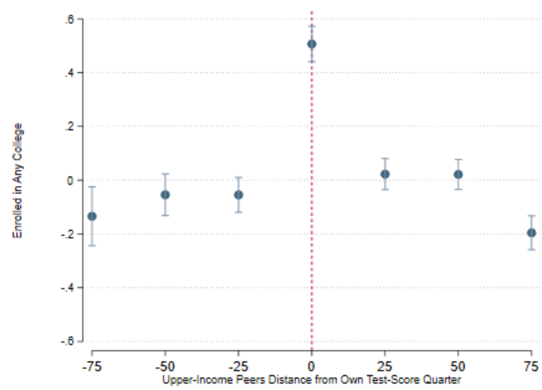
(h) Residual Lead Cohort Wages

Notes: Figures present the residualized outcomes against residualized cohort shares of upper-income students. Residuals are constructed from regressions that include school and cohort fixed effects and control for student demographics and grade 8 test scores, corresponding to the baseline specification (Models (1) in Tables 4 and 5). In Panels (b) and (f), residuals additionally include school-specific linear trends, matching the preferred specification (Model (3) in Tables 4 and 5). Panels (c), (d), (g), and (h) implement the placebo tests by assigning students the upper-income share of adjacent cohorts (lead/lag), with residuals constructed including school trends (Model (5) and (6) in Tables 4 and 5). The sample is limited to lower-income students with grade 8 reading scores. The bins are based on grouping the x-values into 20 equal-sized bins.

Figure 8: Impact of Having a Higher Share of Upper-Income Students by Proximity to Lower-Income Student's Own Test Score



(a) Wage (2023)



(b) Any College Enrollment

Notes: The figure captures the impact of having a higher share of upper-income students who (1) scored in the same quartile of grade 8 reading scores as the focal student (proximity= 0), (2) scored one quartile above (proximity= 25) or below (proximity= −25), (3) scored two quartile above (proximity= 50) or below (proximity= −50), and (4) scored three quartiles above (proximity= 75) or below (proximity= −75). Coefficients are derived from a regression analogous to that in Table 8, with the share of upper- and lower-income peers decomposed by test score proximity. The model controls for the student's own grade 8 reading test score quartile.

8 Supplementary Appendix

8.1 Appendix A: Notes on Simulation Calculations

To calculate the expected exposure to higher-income students for each income group under various random assignment levels, excluding own status, I do the following. I calculate the expected number of (higher-income) peers in each classroom had students been randomly assigned to classrooms in unit j. To calculate the expected number of students in each classroom, I divide the total number of student-classroom enrollments in Texas state public schools in unit j in each year-cohort $TotalEnrollments_j$ by the total number of classrooms offered in unit j (N_j). To calculate the expected number of higher-income students, I calculate the total number of student-classroom enrollments for higher-income students in unit j in each year-cohort $TotalHighIncomeEnrollments_j$ and divide it by the total number of classrooms offered in unit j (H_j). The choice of unit j depends on the random assignment level (state/district/school). This is shown in Equation (2)).

$$\begin{aligned} H_j &= \frac{TotalHighIncomeEnrollments_j}{TotalClassrooms_j} \\ N_j &= \frac{TotalEnrollments_j}{TotalClassrooms_j} \end{aligned} \tag{2}$$

For each student, her expected proportion of higher-income peers had students been randomly assigned to classrooms in unit j is equal to the proportion of higher-income students in unit j, i.e., $\frac{H_j}{N_j}$ adjusted to exclude the student herself. To adjust the proportion to exclude a student's own status, if a student is a higher-income student, I subtract from both the denominator and numerator her number of classroom enrollments n_{ij} during a given year in unit j. If a student is lower-income, I subtract from the denominator only her number of classroom enrollments as shown in Equations (3)) and (4)).

$$ExposureRand_{Hj} = \frac{H_j - n_{ij}}{N_j - n_{ij}} \tag{3}$$

$$ExposureRand_{Lj} = \frac{H_j}{N_j - n_{ij}} \tag{4}$$

To calculate the cumulative expected exposure for a given student across expected grades, I calculate the average of her weighted proportion of higher-income classmates across the years. The weights are based on the number of classrooms a student is enrolled in each year.

8.2 Appendix B: Add Health Data and Cross-Income Friendships

The Add Health at-home survey has information on parental income but only for a random subsample of students from the wider in-school survey. Therefore, I use the socioeconomic variables available for all students to build a model that predicts student income to capture all student SESs who share the same school extracurricular. I use a simple linear regression and use the following variables to predict student income: Father and Mother's occupation, Father and Mother's education, Father and Mother's employment status, number of individuals in the household, and missing indicators. I tested and built this model within the subset of students who were surveyed at home for whom I have income data. Then, I used this model to predict the income for the entire sample of students with the in-school survey. The students were then divided into three income groups based on their predicted income that mimic the income group sizes in the Texas data.

High-SES students are those whose predicted income is in the top 24 percentiles, and low-SES students are students whose predicted income is in the bottom 29 percentiles. These three income indicators seem to effectively capture the variation in parental income. The median income is 26.6, 40.4 and 67.7 thousand dollars for low-, middle-, and high-SES students, respectively.[71] I summarize the characteristics of each of the income groups (including parental education) in Table A1.

To identify the relationship between exposure to upper-income students and individual students' (close) cross-income friendships, I use the Add Health data. The Add Health data contain infor-

71. The median is based on the full sample of students in the at-home survey for whom I have parental income data. I list these in 1994 dollars, which are equivalent to 46.45, 70.5 and 118.21 thousand 2020 dollars. I split the set of students surveyed at home into two equally sized random groups. The in-sample half is used to build the model, and the out-of-sample half is used to test the model. The average income for out-of-sample high-SES students surveyed at home is 68.6 thousand dollars.

mation on individual student friendship patterns and enrollment in extracurricular activities. The in-school survey asks students to list up to five male and five female friends. However, they lack information on which classrooms students are enrolled in, and as such, I use it to capture the relationship between school (and extracurricular activities) peer income composition and friendship formation. Table A2 summarizes the demographics of students in Add Health.[72] The Add Health data list 33 extracurricular activities that students can check. The extracurricular activities listed are the following: yearbook, student council, honor society, newspaper, wrestling, volleyball, track, tennis, swimming, soccer, ice hockey, football, field hockey, basketball, baseball/softball, other sport, orchestra, chorus or choir, cheerleading/dance team, band, science club, math club, history club, Future Farmers of America, drama club, debate team, computer club, book club, Spanish club, Latin club, German club, French club, other club or organization.

To capture the relationship between school (and extracurricular activities) peer income composition and friendship formation, I calculate the proportion of each student's listed friends who are high SES in the Add Health data. I define high-SES students as those in the top 24 percentiles of predicted income. Then, I calculate the proportion of a school's student population that is high-SES and the proportion of a student's peers in extracurricular activities that are high-SES (excluding herself). Last, I run a regression with the proportion of high-SES peers in school (and another with the proportion of high-SES peers in extracurriculars) on the student's proportion of high-SES friends. Relationship is shown in Figure A8.

72. I used the first wave of surveys from 1994 to 1995 that included in-school data for a sample of 85,627 students in 142 schools. I have information on friends and their SES for 77% of the students surveyed in school. For more information on how the Add Health survey was collected, see Harris et al. (2019)).

8.3 Appendix C: Tables and Figures

Table A1: Student School and Classroom Enrollment by Expected Grade

Variable	5	6	7	8	9	10	11	12
Enrolled in Charter School	.037	.057	.058	.06	.054	.058	.057	.053
	(.188)	(.232)	(.235)	(.238)	(.226)	(.235)	(.233)	(.224)
Enrolled in Alternative School	.006	.009	.011	.015	.024	.036	.045	.042
	(.08)	(.094)	(.105)	(.122)	(.153)	(.187)	(.207)	(.2)
Enrolled in Title 1 School	.706	.612	.568	.57	.418	.413	.421	.435
	(.455)	(.487)	(.495)	(.495)	(.493)	(.492)	(.494)	(.496)
Special Education	.089	.091	.09	.089	.087	.085	.083	.083
	(.285)	(.288)	(.286)	(.284)	(.281)	(.279)	(.276)	(.275)
Gifted Program	.094	.097	.099	.099	.096	.095	.098	.105
	(.292)	(.296)	(.298)	(.299)	(.294)	(.294)	(.298)	(.306)
Enrolled in Bilingual Program	.171	.08	.069	.071	.075	.078	.063	.035
	(.377)	(.271)	(.254)	(.256)	(.263)	(.268)	(.244)	(.185)
Number of Advanced Courses Taken	0	0	.001	.026	.332	.69	1.721	1.675
	(0)	(.006)	(.041)	(.237)	(.804)	(1.345)	(2.709)	(2.264)
Number of Classrooms	9.889	9.396	9.70	10.077	14.767	14.861	12.095	8.481
	(3.968)	(3.764)	(3.688)	(3.5)	(3.148)	(3.214)	(4.595)	(4.308)
Number of Courses Taken	9.277	8.632	8.806	9.288	14.167	14.333	11.695	8.232
	(3.615)	(3.32)	(3.248)	(3.112)	(2.823)	(2.868)	(4.179)	(3.906)
Any Vocational Ed	.055	.081	.207	.377	.691	.797	.832	.842
	(.229)	(.273)	(.405)	(.485)	(.462)	(.402)	(.374)	(.365)
Number of Students	1124550	1093343	1076510	1064386	1052391	1035723	988132	905982

This table describes average enrollment across three cohorts of students followed from grade 5 to 12 from 2012 to 2021 in each type of school each year as well as a description of the course offerings in each type of school. The number in brackets is the standard deviation from the mean.

Table A2: Add Health Student Summary Statistics

Variable	Mean
Hispanic	0.1190
White	0.5952
Black	0.2034
Either Parent with College Degree	0.3632
Either Parent in Prof. Occupation	0.4444
Parental Income	43.1524
Median Income	35.0000
Proportion High-SES	0.2416
Proportion Low-SES	0.2872
Number of Students	85627
Number of Schools	142

This table summarizes demographic statistics for students surveyed in the Add Health in-school surveys. The variables "Median Income" and "Parental Income" are based on the at-home surveys of a random subset of students (17,238) for whom there are income data. The high- and low-SES indicators are based on the predicted income measure described in Section 3.2. All averages are weighted by survey sampling weights to be nationally representative.

Table A3: Correlation Between Proportion Upper-Income Students in High School and Lower-Income Students' Demographic Characteristics

	(1)	(2)	(3)	(4)
Black Student	-0.0423	-0.0397	-0.00441	-0.000735
	(0.0155)	(0.0153)	(0.0134)	(0.0134)
Group Mean	0.150	0.150	0.150	0.150
N Clusters	2358	2358	2358	2358
White Student	0.0412	0.0445	0.0195	0.0231
	(0.0118)	(0.0119)	(0.0134)	(0.0134)
Group Mean	0.0900	0.0900	0.0900	0.0900
N Clusters	2358	2358	2358	2358
Hispanic Student	0.00174	-0.00479	-0.00520	-0.0133
	(0.0197)	(0.0196)	(0.0186)	(0.0188)
Group Mean	0.680	0.680	0.680	0.680
N Clusters	2358	2358	2358	2358
Ever Immigrant	-0.0434	-0.0686	-0.00282	-0.0156
	(0.0152)	(0.0155)	(0.0112)	(0.0114)
Group Mean	0.100	0.100	0.100	0.100
N Clusters	2358	2358	2358	2358
Female Student	0.0133	0.0372	-0.00575	0.0118
	(0.0160)	(0.0160)	(0.0193)	(0.0195)
Group Mean	0.470	0.470	0.470	0.470
N Clusters	2358	2358	2358	2358
Income Reported on FAD	-9.341	9.768	4.947	14.67
	(23.26)	(23.74)	(23.64)	(23.67)
Group Mean	60.27	60.27	60.27	60.27
N Clusters	2153	2153	2153	2153
N	505152	505152	505152	505152
Predicted Wage	683.4	962.4	105.6	204.5
	(264.6)	(268.7)	(215.7)	(219.6)
Group Mean	20670.7	20670.7	20670.7	20670.7
N Clusters	2358	2358	2358	2358
Predicted College Enrollment	0.0102	0.0199	-0.000213	0.00522
	(0.00518)	(0.00520)	(0.00462)	(0.00467)
Group Mean	0.370	0.370	0.370	0.370
N Clusters	2358	2358	2358	2358
School time trend			X	X
Proportion High-Achieving Peers		X		X

Similar to Table 3, but controlling for student demographics and test scores, leaving out the outcome we are testing balance on. For the predicted college and wage outcome, I only control for grade 8 test score since the other demographic characteristics were used to predicted wage and college enrollment. Standard errors are in parentheses and clustered at the school level.

Table A4: Impact of Proportion Upper-Income and High-Achieving Students on College Enrollment for Lower-Income Students

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Enrolled in College/University						
Proportion Upper-Income	0.0918		0.127	0.0555		0.0907
	(0.0224)		(0.0226)	(0.0210)		(0.0210)
Proportion High-Achieve		-0.103	-0.117		-0.146	-0.152
		(0.0189)	(0.0187)		(0.0162)	(0.0161)
Group Mean	0.370					
Enrolled in a Public 2yr College						
Proportion Upper-Income	0.125		0.146	0.0460		0.0706
	(0.0238)		(0.0239)	(0.0223)		(0.0228)
Proportion High-Achieve		-0.0536	-0.0702		-0.101	-0.106
		(0.0181)	(0.0178)		(0.0158)	(0.0161)
Group Mean	0.290					
Enrolled in a Public 4yr College						
Proportion Upper-Income	0.0340		0.0561	0.0491		0.0705
	(0.0142)		(0.0139)	(0.0156)		(0.0155)
Proportion High-Achieve		-0.0674	-0.0738		-0.0876	-0.0927
		(0.0137)	(0.0135)		(0.0126)	(0.0125)
Group Mean	0.140					
Enrolled in a Selective University						
Proportion Upper-Income	0.00752		0.0131	0.00699		0.0125
	(0.00397)		(0.00412)	(0.00470)		(0.00482)
Proportion High-Achieve		-0.0170	-0.0185		-0.0228	-0.0237
		(0.00358)	(0.00363)		(0.00444)	(0.00448)
Group Mean	0.0200					
School Time Trends				X	X	X
N	1501742	1501742	1501742	1501742	1501742	1501742
N Clusters	2358	2358	2358	2358	2358	2358

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). Models (1)–(3) do not include school time trends. College enrollment outcomes include cohorts 2010–2019. Models (3) and (6) include both the proportion of upper-income and higher-achieving peers in the cohort. Group Mean is the average for lower-income students in the sample.

Table A5: Impact of Proportion Upper-Income and High-Achieving Students on College Graduation and Wages for Lower-Income Students

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Graduated College/University						
Proportion Upper-Income	0.0541		0.0605	0.0249		0.0311
	(0.0213)		(0.0211)	(0.0229)		(0.0229)
Proportion High-Achieve		-0.0148	-0.0222		-0.0163	-0.0194
		(0.0191)	(0.0191)		(0.0175)	(0.0177)
Group Mean	0.170					
Wage 2023						
Proportion Upper-Income	3190.6		3700.9	4337.8		5141.6
	(1409.3)		(1425.5)	(1670.4)		(1689.0)
Proportion High-Achieve		-1320.7	-1774.2		-2938.7	-3451.1
		(985.0)	(987.6)		(1079.5)	(1084.9)
Group Mean	20490.5					
Income Percentile 2023						
Proportion Upper-Income	1.879		3.020	3.560		4.412
	(1.363)		(1.383)	(1.620)		(1.638)
Proportion High-Achieve		-3.597	-3.967		-3.315	-3.778
		(1.009)	(1.013)		(1.062)	(1.064)
Group Mean	46.75					
School Time Trends				X	X	X
N	700245	700245	700245	700245	700245	700245
N Clusters	1992	1992	1992	1992	1992	1992

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). Models (3) and (6) include both the proportion of upper-income and higher-achieving peers in the cohort. College graduation and wages include cohorts 2010–2014. Group Mean is the average for lower-income students in the sample.

Table A6: Robustness To Model Specifications (School Trend Linearity Assumption): College Enrollment Outcomes for Lower-Income Students

	Model 1	Model 2	Model 3	Model 4	Model 5
Enrolled in College/University					
Proportion Upper-Income	0.0467	0.0503	0.0210	0.0686	0.0415
	(0.0225)	(0.0273)	(0.0208)	(0.0207)	(0.0208)
Group Mean	0.360	0.360	0.370	0.370	0.370
Enrolled in a Public 2yr College					
Proportion Upper-Income	0.0484	0.0609	0.0232	0.0720	0.0518
	(0.0232)	(0.0279)	(0.0218)	(0.0211)	(0.0218)
Group Mean	0.290	0.280	0.290	0.290	0.290
Enrolled in a Public 4yr College					
Proportion Upper-Income	0.0387	0.0322	0.0206	0.0494	0.0215
	(0.0154)	(0.0181)	(0.0155)	(0.0152)	(0.0156)
Group Mean	0.140	0.130	0.140	0.140	0.140
Enrolled in Selective University					
Proportion Upper-Income	0.00150	0.0111	0.00285	0.0100	0.00368
	(0.00498)	(0.00508)	(0.00493)	(0.00469)	(0.00496)
Group Mean	0.0200	0.0100	0.0200	0.0200	0.0200
School time trend	X				
Quadratic School trend			X		
N clusters	2146	1705	2358	2353	2347
N	1221309	671294	1501742	1490035	1480522

Standard errors are in parentheses and clustered at the school level. Model (1) re-estimating the main specifications based on Equation (1) restricting to schools whose residual standard deviation lies within the school-specific 95% interval of likely deviations from the school mean based on a simulated distribution of deviations from the school mean discussed in more details in Section 5.2. Model (2) excludes school trends and only includes schools whose cohort share upper-income students' deviation from school mean lies within the school-specific 95% interval of likely deviations based on a simulated distribution of deviations from the school mean. Model (3) includes a quadratic school trend for each school. Model (4) excludes school trends and instead bins each schools' first and second five years. Model (5) excludes school trends and bins for each school the first six years into two three-year bin and the last four years into one bin in Model. Model (4) and (5) exclude school-cohort bins with fewer than three cohorts.

Table A7: Robustness To Model Specifications (School Trend Linearity Assumption): College Graduation and Wages for Lower-Income Students

	Model 1	Model 2	Model 3
Graduated College/University			
Proportion Upper-Income	0.0249	0.0380	0.00974
	(0.0258)	(0.0278)	(0.0295)
Group Mean	0.130	0.130	0.130
Wage 2023			
Proportion Upper-Income	3728.9	3736.4	3569.7
	(1958.1)	(1979.0)	(2005.0)
Group Mean	17122.8	16739.8	17076.6
Income Percentile 2023			
Proportion Upper-Income	3.803	4.363	2.850
	(1.877)	(1.936)	(2.003)
Group Mean	48.78	48.29	48.61
School time trend	X		
Quadratic School trend			X
N clusters	1795	1379	1992
N	564484	306626	700245

Standard errors are in parentheses and clustered at the school level. Model (1) re-estimating the main specifications based on Equation (1) restricting to schools whose residual standard deviation lies within the school-specific 95% interval of likely deviations from the school mean based on a simulated distribution of deviations from the school mean discussed in more details in Section 5.2. Model (2) excludes school trends and only includes schools whose cohort share upper-income students' deviation from school mean lies within the school-specific 95% interval of likely deviations based on a simulated distribution of deviations from the school mean. Model (3) includes a quadratic school trend for each school.

Table A8: Impact of Proportion Upper-Income Students on Lower-Income Students' College Enrollment and wages: Robustness to Specification

	Model 1	Model 2	Model 3	Model 4	Model 5
Enrolled in College/University					
Proportion Upper-Income	0.143	0.0783	0.0787	0.0555	0.0585
	(0.0248)	(0.0209)	(0.0205)	(0.0210)	(0.0229)
Group Mean	0.370	0.370	0.370	0.370	0.370
N Clusters	2358	2358	2358	2358	2355
Enrolled in a Public 2yr College					
Proportion Upper-Income	0.164	0.0630	0.0630	0.0460	0.0310
	(0.0253)	(0.0221)	(0.0218)	(0.0223)	(0.0254)
Group Mean	0.290	0.290	0.290	0.290	0.290
N Clusters	2358	2358	2358	2358	2355
Enrolled in a Public 4yr College					
Proportion Upper-Income	0.0634	0.0625	0.0630	0.0491	0.0480
	(0.0153)	(0.0153)	(0.0151)	(0.0156)	(0.0169)
Group Mean	0.140	0.140	0.140	0.140	0.140
N Clusters	2358	2358	2358	2358	2355
Enrolled in a Selective University					
Proportion Upper-Income	0.0121	0.00900	0.00916	0.00699	0.00903
	(0.00400)	(0.00470)	(0.00470)	(0.00470)	(0.00577)
Group Mean	0.0200	0.0200	0.0200	0.0200	0.0200
N Clusters	2358	2358	2358	2358	2355
Graduated College/University					
Proportion Upper-Income	0.0612	0.0394	0.0385	0.0249	0.0350
	(0.0208)	(0.0227)	(0.0226)	(0.0229)	(0.0276)
Group Mean	0.170	0.170	0.170	0.170	0.170
N Clusters	1992	1992	1992	1992	1991
Wage 2023					
Proportion Upper-Income	4076.4	5304.4	5289.4	4335.1	3751.5
	(1500.9)	(1732.4)	(1735.5)	(1669.9)	(2245.7)
Group Mean	20490.5	20490.5	20490.5	20490.5	20490.5
N Clusters	1992	1992	1992	1992	1991
Income Percentile 2023					
Proportion Upper-Income	2.966	4.567	4.586	3.556	3.461
	(1.444)	(1.700)	(1.704)	(1.619)	(2.233)
Group Mean	46.75	46.75	46.75	46.75	46.75
N Clusters	1992	1992	1992	1992	1991
Linear School Time Trends		X	X	X	X
Student Demographic Controls			X	X	X
Student G8 Test Controls				X	X
Grade 8 School-Cohort Fixed Effects					X
Proportion High-Achieving Students					

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). All models include cohort and school fixed effects. College enrollment outcomes include cohorts 2010–2019, and graduation and wages outcomes include cohorts 2010–2014. Group Mean is the average for lower-income students in the sample. Student demographic controls include student own race, immigrant status and gender. Student grade 8 test controls include their grade 8 math and reading scores and an indicator for missing grade 8 math and reading tests.

Table A9: Examining the Impact on Lower-Income Students' College Enrollment

	(1) FAD	(2) Any Public College	(3) Only 2yr Public	(4) Only 4-Year Public	(5) Both 2yr and 4yr Public
Proportion Upper-Income Students	0.0528	0.0568	0.00768	0.0108	0.0384
	(0.0197)	(0.0208)	(0.0175)	(0.0116)	(0.0137)
N Clusters	2358	2358	2358	2358	2358
N	1501742	1501742	1501742	1501742	1501742

Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1) and include cohorts 2010–2019.

Table A10: Impact of Increase in Proportion Higher-Income Students: Interacted with Drive Time to Nearest 4-year College

	(1) Any College	(2) 2yr Public	(3) 4yr Public	(4) Selective	(5) Grad Any
Proportion Upper-Income Students	0.0997	0.0768	0.0763	0.0233	0.0978
	(0.0425)	(0.0464)	(0.0330)	(0.00898)	(0.0414)
Proportion Upper-Income Students× Drive Time to closest 4-Year College	-0.00130	-0.000938	-0.000740	-0.000438	-0.00188
	(0.000792)	(0.000829)	(0.000618)	(0.000153)	(0.000871)
N Clusters	2231	2231	2231	2231	1911
N	1415088	1415088	1415088	1415088	654254

Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1), but additionally including an interaction with a students' high school distance to the nearest 4yr college based on Acton, Cortes, Miller, and Morales (2025) calculation. It include cohorts 2010–2019.

Table A11: Impact of Increase in Proportion Higher-Income Students: Test for Nonlinearity

	(1) Any College	(2) 2yr Public	(3) 4yr Public	(4) Selective	(5) Grad Any	(6) Wage 2023	(7) Income Percentile
Proportion Upper-Income Students	0.100	0.114	0.0835	-0.00181	-0.00850	5643.5	2.597
	(0.0432)	(0.0467)	(0.0329)	(0.00951)	(0.0522)	(3442.0)	(3.323)
Proportion Upper-Income Students^2	-0.0873	-0.132	-0.0671	0.0174	0.0661	-2482.1	1.805
	(0.0650)	(0.0692)	(0.0508)	(0.0176)	(0.0844)	(5816.7)	(5.465)
N Clusters	2358	2358	2358	2358	1992	1992	1992
N	1501742	1501742	1501742	1501742	700245	700245	700245

Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1), but additionally including an interaction with a students' high school distance to the nearest 4yr college based on Acton, Cortes, Miller, and Morales (2025) calculation. It include cohorts 2010–2019.

Table A12: Impact of Increase in Proportion Higher-Income Students on Returns to Major and Peer Income at College

	(1) Institution Peer Avg Income	(2) Expected Major Return (25-34)	(3) Expected Major Return (35-44)
Proportion Upper-Income Students	94719.9	1179.2	2438.7
	(53350.7)	(1479.3)	(1966.6)
Group Mean	544563.4	56695.4	77202.5
N Clusters	2166	1669	1669
N	556164	120870	120870

Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1) Model (1) include cohorts 2010–2019 and only include lower-income students who enroll in college. The estimated institution peer average income is based on the average reported parental income of students enrolled in the institution. For students who enroll in multiple colleges, I assign them the institution with the highest peer average income. Models (2)-(3), includes cohorts 2010-2014 and only students who graduated college. A students' major is based on their reported major at graduation linked to ACS field expected income.

Table A13: Impact of Proportion Upper-Income Conditional on Peer Achievement: Sensitivity to Measure of Achievement

	Model 1	Model 2	Model 3	Model 4	Model 5
Enrolled in College/University					
Proportion Upper-Income	0.0639	0.0833	0.0947	0.0459	0.0759
	(0.0211)	(0.0206)	(0.0211)	(0.0194)	(0.0207)
Group Mean	0.370				
Enrolled in a Public 2yr College					
Proportion Upper-Income	0.0548	0.0666	0.0720	0.0380	0.0594
	(0.0222)	(0.0220)	(0.0230)	(0.0213)	(0.0224)
Group Mean	0.290				
Enrolled in a Public 4yr College					
Proportion Upper-Income	0.0513	0.0655	0.0757	0.0391	0.0627
	(0.0157)	(0.0154)	(0.0156)	(0.0142)	(0.0154)
Group Mean	0.140				
Enrolled in a Selective University					
Proportion Upper-Income	0.00728	0.0103	0.0140	0.00731	0.00958
	(0.00476)	(0.00473)	(0.00486)	(0.00464)	(0.00475)
Group Mean	0.0200				
Graduated College/University					
Proportion Upper-Income	0.0460	0.0474	0.0306	0.0336	0.0325
	(0.0226)	(0.0225)	(0.0230)	(0.0227)	(0.0229)
Group Mean	0.170				
Average Quarterly Wage 2023					
Proportion Upper-Income	5318.9	5502.0	5258.4	5422.3	4742.6
	(1669.2)	(1669.5)	(1690.2)	(1703.2)	(1678.8)
Group Mean	20490.5				
Income Percentile 2023					
Proportion Upper-Income	4.268	4.513	4.481	4.678	4.073
	(1.618)	(1.624)	(1.641)	(1.655)	(1.629)
Group Mean	46.75				
Proportion High-Achieving (Math G8)	X				
Proportion High-Achieving (Math+Reading G8)		X			
Proportion High-Achieving (Reading G3-G8)			X		
Proportion High-Achieving (Reading G3)				X	
Peer Average Std. Test Score (Reading G8)					X

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1). College enrollment outcomes include cohorts 2010–2019 (N: 1501742; N Clusters: 2358). College graduation and wages include cohorts 2010–2014 (N: 700245; N Clusters: 1992). All models include both a peer achievement control. The measure of achievement varies across models. Models (1)–(4) include a control for the proportion high-achieving peers where high-achieving students are those who score in the top quarter of the standardized test score distribution. Model (4) controls for students' own grade 3 reading and math test scores instead of grade 8. Model (5) controls for a continuous measure of average grade 8 reading scores for all students. Group Mean is the average for lower-income students in the sample.

Table A14: Impact of Share Upper-Income Students in Cohort on Classroom Exposure

	(1) Classroom Exposure	(2) AP Courses	(3) Advanced Courses
Proportion Upper-Income and High-Achieve	0.569	-0.326	-0.566
	(0.00836)	(0.242)	(0.352)
Proportion Upper-Income and Lower-Achieve	0.607	0.954	1.258
	(0.00835)	(0.228)	(0.306)
N	1369434	1501742	1501742
Group Mean	0.130	1.280	2.270
N Clusters	2322	2358	2358

Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1) but instead of the main independent variable (proportion upper-income students), upper-income peers are split into two groups based on if scored in the top quartile of their grade 8 reading test (Proportion Upper-Income and High-Achieve) or not (Proportion Upper-Income and Lower-Achieve). Classroom exposure is only available for cohorts 2011 onward. Models (2) and (3) include cohorts 2010–2014.

Table A15: Impact of Share Upper-Income Students in Cohort on College Enrollment

	(1) White Students	(2) Black Students	(3) Hispanic Students
Enrolled in Any College			
Upper-Income, Same Race/Ethnicity	0.0621	0.0785	0.122
	(0.0408)	(0.166)	(0.0655)
Upper-Income, Different Race/Ethnicity	-0.0247	0.103	0.0107
	(0.0845)	(0.0562)	(0.0321)
Group Mean	0.310	0.380	0.390
Enrolled in a Public 2yr			
Upper-Income, Same Race/Ethnicity	0.0514	-0.0829	0.0953
	(0.0399)	(0.156)	(0.0781)
Upper-Income, Different Race/Ethnicity	-0.0372	0.110	0.0101
	(0.0827)	(0.0542)	(0.0328)
Group Mean	0.260	0.270	0.310
Enrolled in a Public 4yr			
Upper-Income, Same Race/Ethnicity	0.0327	0.486	0.0878
	(0.0259)	(0.125)	(0.0523)
Upper-Income, Different Race/Ethnicity	-0.0414	0.0867	0.0160
	(0.0565)	(0.0427)	(0.0229)
Group Mean	0.100	0.150	0.150
Highly Selective University			
Upper-Income, Same Race/Ethnicity	0.00538	0.0658	0.0203
	(0.00984)	(0.0389)	(0.0139)
Upper-Income, Different Race/Ethnicity	-0.00852	-0.00836	-0.000124
	(0.0216)	(0.0134)	(0.00730)
Group Mean	0.0100	0.0100	0.0200
N	131812	228839	1017112
N Clusters	1980	1756	2220

Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1) but instead of the main independent variable (proportion upper-income students), upper-income peers are split into two groups: Upper-income students of the same race/ethnicity and upper-income students of different race/ethnicity. Each model presents the estimates for each racial/ethnic group separately: for lower-income White, Black and Hispanic students in Models (1), (2) and (3), respectively. All outcomes include cohorts 2010–2019.

Table A16: Impact of Share Upper-Income Students in Cohort on Graduating College and Wages

	(1) White Students	(2) Black Students	(3) Hispanic Students
Graduated College/University			
Upper-Income, Same Race/Ethnicity	0.0164	0.211	0.0406
	(0.0471)	(0.178)	(0.0690)
Upper-Income, Different Race/Ethnicity	0.0179	-0.0287	0.0268
	(0.105)	(0.0684)	(0.0386)
Group Mean	0.130	0.140	0.200
Wage (2023)			
Upper-Income, Same Race/Ethnicity	10957.2	3688.8	6148.1
	(3583.2)	(14226.3)	(4141.7)
Upper-Income, Different Race/Ethnicity	-6350.8	11007.1	-260.7
	(8118.3)	(5037.5)	(2796.8)
Group Mean	19774.6	18435.0	21695.4
Income Percentile			
Upper-Income, Same Race/Ethnicity	14.48	-7.832	4.460
	(3.637)	(10.87)	(4.014)
Upper-Income, Different Race/Ethnicity	-1.906	7.201	-1.530
	(7.786)	(4.492)	(2.708)
Group Mean	45.82	46.87	47.58
N	60843	106733	475618
N Clusters	1668	1390	1885

Similar to Table A15, but looking at long-term outcomes. For longer-term outcomes, sample is limited to cohorts 2010–2014.

Table A17: Impact of Share Upper-Income Students in Cohort on the Share of Upper-Income Classmates

	(1) White Students	(2) Black Students	(3) Hispanic Students
Upper-Income, Same Race/Ethnicity	0.620	0.666	0.668
	(0.00989)	(0.0400)	(0.0151)
Upper-Income, Different Race/Ethnicity	0.588	0.547	0.579
	(0.0154)	(0.0140)	(0.0102)
Group Mean	0.250	0.150	0.110
N Clusters	1952	1710	2186
N	120196	208360	926030

Similar to Table A15, but looking at classroom exposure defined as the share of total classmates (in the same classroom) in grade 9 who are upper-income. Classroom data is only available starting 2011, so the sample is limited to cohorts 2011–2019.

Table A18: Impact of Changes in the Share of White Peers and Peers with Disciplinary Records in Grade 9 on Lower-Income Students' College Enrollment

	Model 1	Model 2	Model 3	Model 4
Enrolled in Any College				
Proportion Upper-Income Students		0.0529		0.0490
		(0.0222)		(0.0205)
Proportion White Students	0.0228	0.00841		
	(0.0165)	(0.0175)		
Proportion with Any Disciplinary Record in G8			0.0141	0.0175
			(0.0153)	(0.0154)
Group Mean	0.370	0.370	0.370	0.370
Enrolled in a Public 2yr				
Proportion Upper-Income Students		0.0418		0.0428
		(0.0237)		(0.0220)
Proportion White Students	0.0247	0.0134		
	(0.0167)	(0.0178)		
Proportion with Any Disciplinary Record in G8			0.0174	0.0204
			(0.0156)	(0.0157)
Group Mean	0.290	0.290	0.290	0.290
Enrolled in a Public 4yr				
Proportion Upper-Income Students		0.0507		0.0439
		(0.0164)		(0.0152)
Proportion White Students	0.00893	-0.00490		
	(0.0113)	(0.0120)		
Proportion with Any Disciplinary Record in G8			0.00173	0.00478
			(0.00962)	(0.00956)
Group Mean	0.140	0.140	0.140	0.140
Highly Selective University				
Proportion Upper-Income Students		0.00751		0.00617
		(0.00483)		(0.00471)
Proportion White Students	0.000631	-0.00141		
	(0.00379)	(0.00387)		
Proportion with Any Disciplinary Record in G8			-0.0000279	0.000400
			(0.00248)	(0.00250)
Group Mean	0.0200	0.0200	0.0200	0.0200
N	1501742	1501742	1501742	1501742
N Clusters	2358	2358	2358	2358

Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1), but looking at the share of White peers alone in Model (1), share of students with any in-school or out-of-school suspensions in grade 8 alone in Model (3). Models (2) and (4) additionally controlling for the share of upper-income peers.

Table A19: Impact of Changes in the Share of White Peers and Peers with Disciplinary Records in Grade 9 on Lower-Income Students' Graduation and Wages

	Model 1	Model 2	Model 3	Model 4
Graduated College/University				
Proportion Upper-Income Students		0.0367		0.0287
		(0.0239)		(0.0226)
Proportion White Students	-0.0201	-0.0303		
	(0.0197)	(0.0207)		
Proportion with Any Disciplinary Record in G8			0.0144	0.0162
			(0.0134)	(0.0134)
Group Mean	0.170	0.170	0.170	0.170
Wage (2023)				
Proportion Upper-Income Students		5020.5		4286.5
		(1734.8)		(1653.4)
Proportion White Students	-637.1	-2069.4		
	(1543.2)	(1599.2)		
Proportion with Any Disciplinary Record in G8			-263.4	12.50
			(968.3)	(963.1)
Group Mean	20490.5	20490.5	20490.5	20490.5
Income Percentile				
Proportion Upper-Income Students		4.242		3.547
		(1.681)		(1.618)
Proportion White Students	-0.892	-2.119		
	(1.508)	(1.562)		
Proportion with Any Disciplinary Record in G8			0.0128	0.241
			(0.916)	(0.918)
Group Mean	46.75	46.75	46.75	46.75
N	700245	700245	700245	700245
N Clusters	1992	1992	1992	1992

Similar to Table A18, but looking at longer-term outcomes: graduation and wages. The sample is limited to cohorts for whom we can observe these outcomes for in early adulthood: cohorts 2010–2014.

Table A20: Impact of Proportion of Upper-Income, Holding Constant Proportion Mid-Income Students, on Lower-Income Students

	(1) Any College	(2) 2yr Public	(3) 4yr Public	(4) Selective	(5) Grad Any	(6) Wage 2023	(7) Income Percentile
Proportion Upper-Income	0.0198	0.0109	0.0312	0.00567	0.0248	5483.6	5.291
	(0.0230)	(0.0240)	(0.0169)	(0.00501)	(0.0253)	(1773.8)	(1.730)
Proportion Middle-Income	-0.0657	-0.0646	-0.0330	-0.00254	-0.00338	2059.4	3.170
	(0.0195)	(0.0200)	(0.0111)	(0.00310)	(0.0176)	(1095.8)	(1.106)
Group Mean	0.370	0.290	0.140	0.0200	0.170	20490.5	46.75
N Clusters	2358	2358	2358	2358	1992	1992	1992
N	1501742	1501742	1501742	1501742	700245	700245	700245

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1) including a control for the share of middle income students in the cohort. The sample is limited to lower-income students. Models (1)–(4) include cohorts 2010–2019 and Models (5)–(7) include cohorts 2010–2014.

Table A21: Impact of Proportion of Upper-Income on Lower-Income Students by Schools' Average Share Upper-Income Students

	(1) Any College	(2) 2yr Public	(3) 4yr Public	(4) Selective	(5) Grad Any	(6) Wage 2023	(7) Income Percentile
Proportion Upper-Income	0.0946	0.0973	0.0783	0.000115	0.00737	5186.6	2.964
	(0.0419)	(0.0452)	(0.0333)	(0.00973)	(0.0487)	(3253.7)	(3.088)
Proportion Upper-Income×Average Share Upper-Income	-0.157	-0.205	-0.117	0.0278	0.0741	-3469.2	2.341
	(0.129)	(0.137)	(0.107)	(0.0372)	(0.166)	(11795.8)	(10.72)
N Clusters	2358	2358	2358	2358	1992	1992	1992
N	1501742	1501742	1501742	1501742	700245	700245	700245

Standard errors are in parentheses and clustered at the school level. Coefficients are based on Equation (1) including an interaction with schools' average share of upper-income students during the years we observe them in. Models (1)–(4) include cohorts 2010–2019 and Models (5)–(7) include cohorts 2010–2014.

Table A22: Bounding the Estimated Impact of the Share of Upper-Income Peers on Wage in 2023 Using Between School Variation

	Model 1	Model 2	Model 3
Proportion Upper-Income	4130.8	5182.8	5171.2
	(500.3)	(453.7)	(611.5)
Group Mean	20490.5	20490.5	20490.5
N Clusters	1995	1994	1995
N	700248	699831	700248
Cohort Fixed Effect	X	X	X
G8 School-Cohort Fixed Effect		X	
District Fixed Effect			X
School Fixed Effect			

Standard errors are in parentheses and clustered at the school level. Models include cohorts 2010-2014. Income for students without unemployment insurance data are imputed with 0. All models include cohort fixed effects and control for student grade 8 test score and demographic characteristics. Model (2) include school grade 8 and cohort in grade 8 school fixed effects based on students' first school in grade 8 enrollment prior to enrolling grade 9. Model (3) includes district fixed effects.

Table A23: Proportion Upper-Income Peers and Lower-Income Students' Average Cohort Income

	Prop FAD	Parent Income (FAD)	Years FRPL	Prop FRPL
Proportion Upper-Income	0.252	112.1	-6.695	-0.795
	(0.0165)	(20.65)	(0.105)	(0.0109)
Group Mean	0.380	94.24	6.400	6.400
N	1501742	1496651	1501742	1501742
N clusters	2358	2271	2358	2358

Standard errors clustered at the school level in parentheses. Each coefficient is based on a regression of the outcome on the proportion upper-income students, including school, cohort fixed effects and school time trends. The outcomes capture a lower-income students' average cohort characteristics excluding own status. "Prop FAD" captures the proportion of a student's cohort who submitted their financial aid (FAD) applications. "Parent Income (FAD)" is the average cohorts' income based on students' reported parental income in their FAD application. Years FRPL captures a student's peers average number of years on free/reduced lunch and prop. FRPL captures a student's peers average proportion of years on free/reduced lunch.

Table A24: Impact of Increase in Proportion Upper-Income Students on Lower-Income Students: Income Defined by Free/Reduced Lunch Status Before High School

	Any College	2yr Public	4yr Public	Selective	Grad Any	Wage 2023	Income Percentile
Proportion of Upper-Income Students	0.0665	0.0543	0.0462	0.0150	0.0279	2219.5	2.122
	(0.0180)	(0.0195)	(0.0122)	(0.00390)	(0.0167)	(1174.6)	(1.160)
Group Mean	0.380	0.300	0.150	0.0200	0.170	20634.2	47.96
N clusters	2358	2358	2358	2358	1992	1992	1992
N	1940788	1940788	1940788	1940788	1097186	1097186	1097186

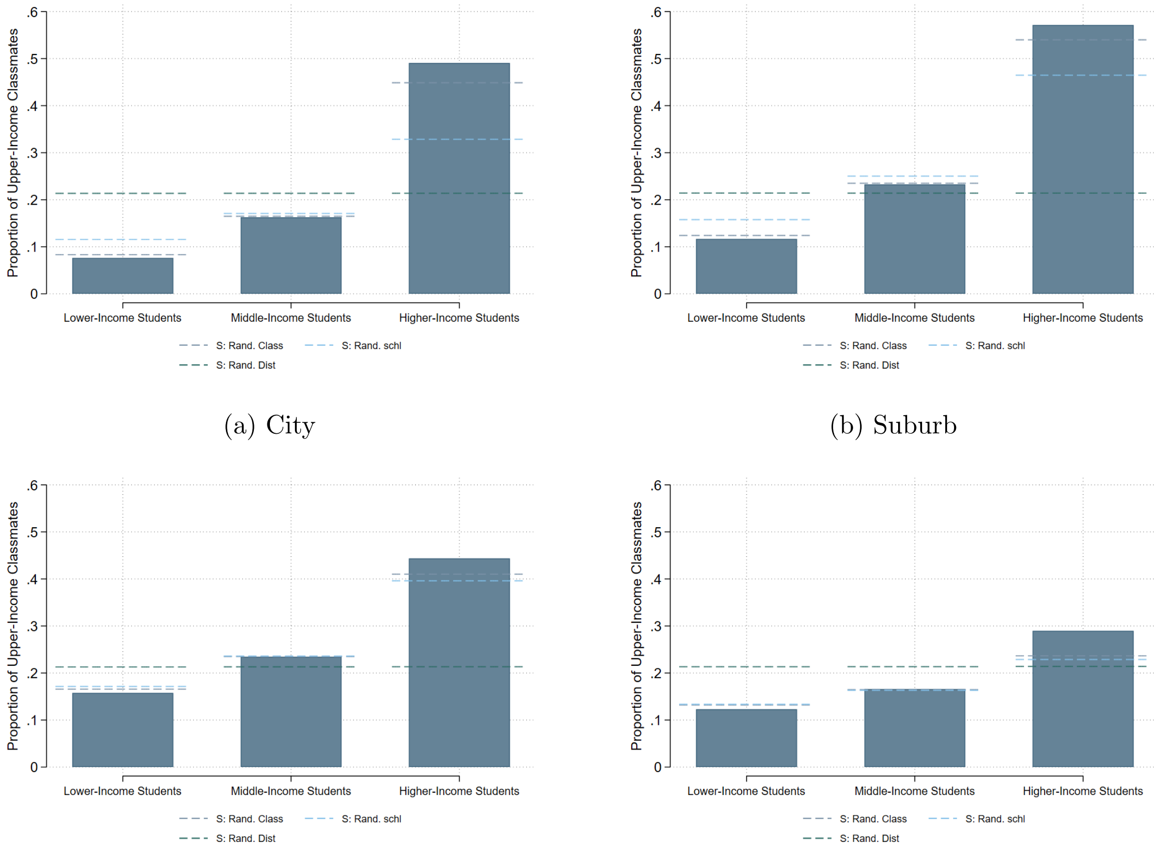
Standard errors are in parentheses and clustered at the school level. The coefficients are based on Equation (1) but with a slightly different definition of income. Lower and upper-income students are defined as students who are always and never on free/reduced lunch status before high school. Models (1)–(4) include cohorts 2010–2019 and Models (5)–(7) include cohorts 2010–2014.

Table A25: Impact of Increase in Share Higher-Income Students on Lower-Income Students' Likelihood of Leaving Initial School

	(1) Leave School	(2) Switch	(3) Attrition
Impact on Lower-Income Students			
Proportion Upper-Income Peers	-0.0640	-0.0232	-0.0408
	(0.0316)	(0.0286)	(0.0145)
Group Mean	0.293	0.190	0.110
N Clusters	2358	2358	2358
Impact on Upper-Income Students			
Proportion Lower-Income	-0.114	-0.116	0.00180
	(0.0400)	(0.0392)	(0.00996)
Group Mean	0.160	0.110	0.0500
N Clusters	2129	2129	2129

Standard errors clustered at the school level in parentheses. For lower-income students (first rows), each coefficient is based on a regression of the outcome on the proportion upper-income peers, including school, cohort fixed effects and school time trends. For upper-income students (second set of rows), each coefficient is based on a regression of the outcome on the proportion lower-income peers, including school, cohort fixed effects and school time trends. The outcomes capture a students likelihood of (1) leaving a given school they are enrolled in grade 9 (either switching schools or attrition), (2) switching schools between grades 9 and 10, and (2) attrition between grades 9 and 10 if we no longer observe them in the data the year after grade 9 either because they dropped out or left the Texas public school system to another private or out-of-state school.

Figure A1: Exposure to Higher-Income Students by District Type: Observed Relative to Random Assignment



(a) City

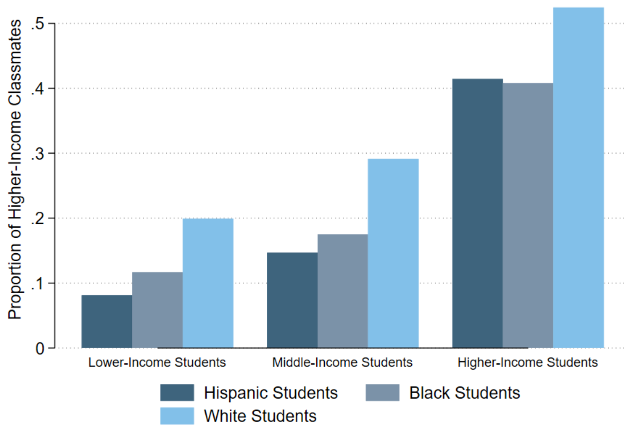
(b) Suburb

(c) Rural

(d) Town

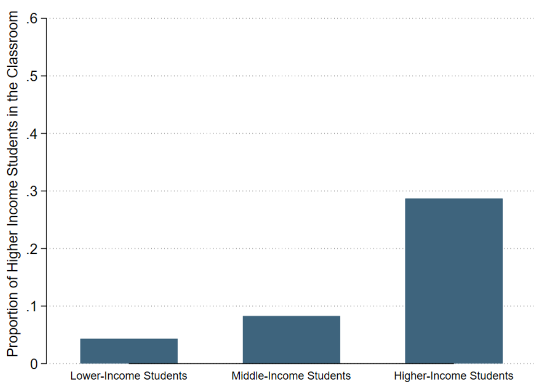
Notes: Similar to Figure 2 but split students by district type. Students are assigned district type based on the district they enroll in grade 5. District type is based on NCES categorization of districts.

Figure A2: Gap in Exposure to Higher-Income Students Exists Independent of Students' Race/Ethnicity

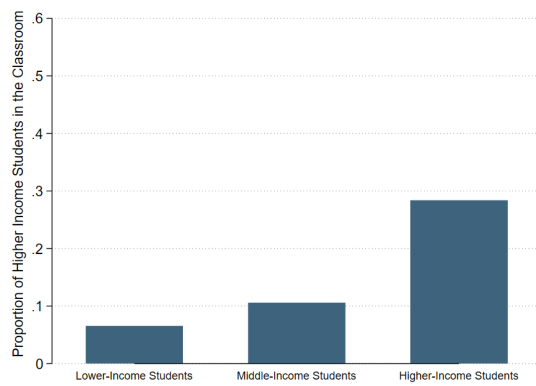


Notes: This plot captures the proportion of higher-income classmates across grades 5 to 12 for each income and racial/ethnic group. Higher-income students are defined as students never on free/reduced lunch.

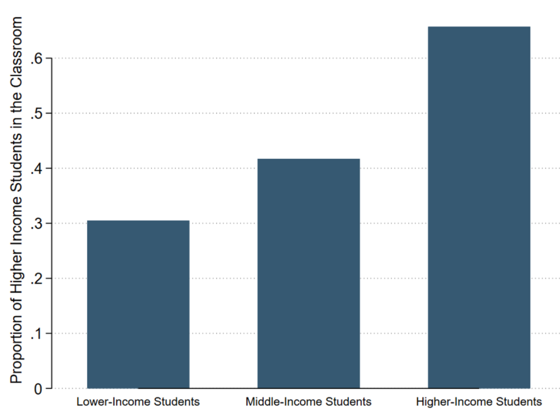
Figure A3: Proportion of Higher-Income, Same-Race/Ethnicity Classmates



(a) Hispanic Students



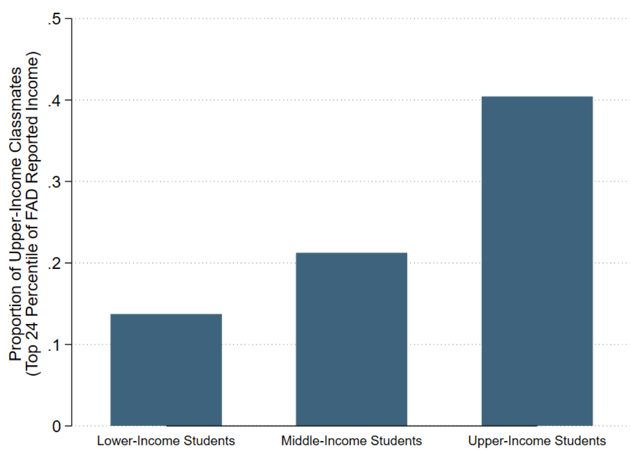
(b) Black Students



(c) White Students

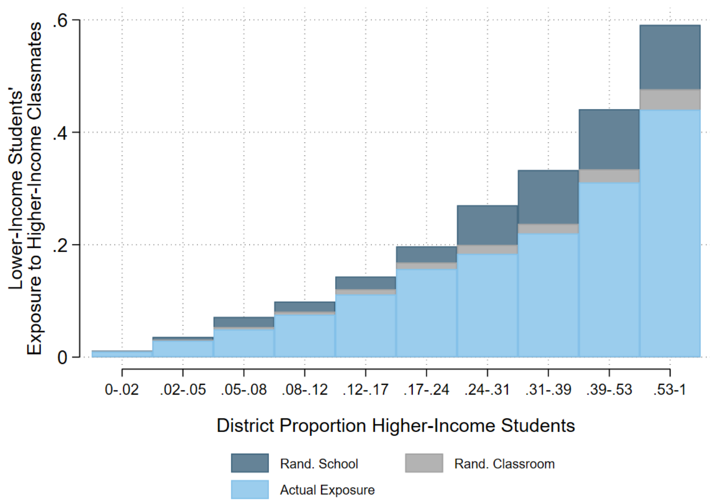
Notes: These plots capture the proportion of higher-income classmates of the same race/ethnicity across grades 5 to 12 for each income and racial/ethnic group. The average is weighted by the number of students across three cohorts in each racial/ethnic group.

Figure A4: Exposure to Higher-Income students is Defined by Parental Income Reported in their Financial Aid Application



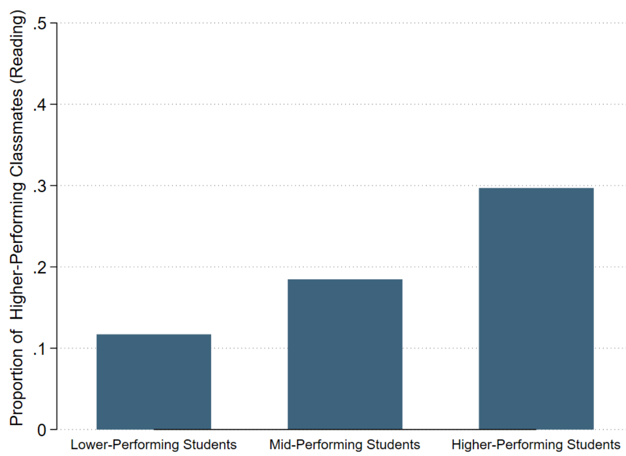
Notes: Parental income is based on parental gross adjusted income listed on the financial aid application. There may be students with parental income equivalent to this or higher but who have not applied for financial aid. Exposure to higher-income students is calculated by dividing the total number a student's classmates between grades 5 and 12 with parental income in the top 24 percentiles based on the distribution of parental income in the financial aid data is divided by the total number of classmates with financial aid data. Lower- middle- and upper-income students on the x-axis are defined by proportion of years in free/reduced lunch status.

Figure A5: The Percentage Point Gap in Exposure to Higher-Income Students Increases as the Proportion of Higher-Income students in the District Increases



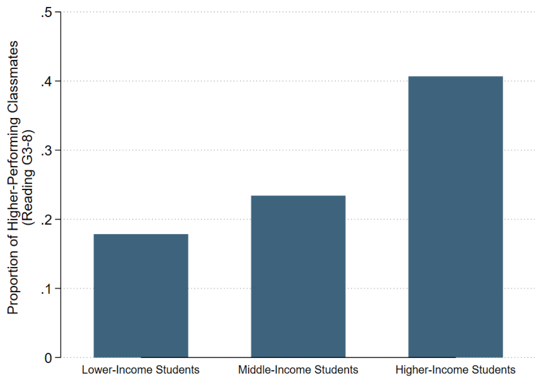
Notes: This plot presents lower-income students' exposure to higher-income students in districts with various proportions of higher-income students. Exposure is defined as the average proportion of higher-income classmates in a year. The light blue bar shows the observed proportion of higher-income students in lower-income students' classrooms. The navy bar presents the expected proportion of higher-income students had students been randomly assigned to schools in the district: school integration benchmark. The gray bar presents the expected proportion of higher-income students had students been randomly assigned to classrooms in the school: classroom integration benchmark. Districts are split into ten percentiles based on the distribution of the proportion of higher-income students in the district in a given school-year. The distribution is weighted by the number of students in each district (independent of income status). The lower and upper bound for each of the percentiles is shown on the x-axis.

Figure A6: The Cross-test score Exposure Gap is Smaller than the Cross-Income Exposure Gap

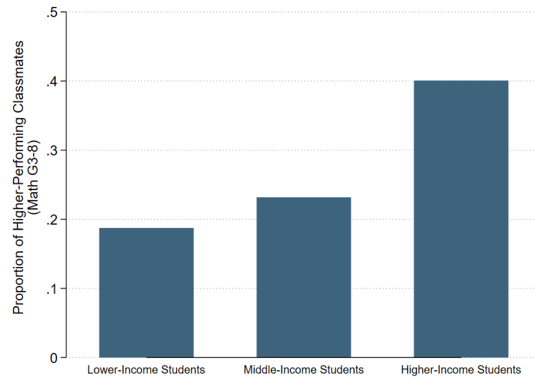


*Notes:*This plot captures students' cumulative proportion of higher-performing classmates between grade 5 and expected grade 12. I define higher-performing students as students who performed in the top 24 percentiles of the test score based on the grade 4 standardized reading test. The test score is standardized based on the raw score for all students who have taken the test in a given year. The percentile is based on the distribution of students who have a grade 4 test score. Approximately 5% of students are missing grade 4 reading test scores. I impute students' missing test score with 0 (i.e., the average test score because the test score is standardized). Lower-achieving students are those who performed in the bottom 29 percentiles of grade 4 test scores.

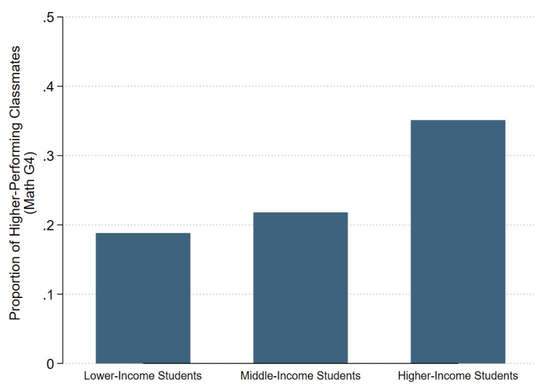
Figure A7: Consistent Gap in Share of Higher-Achieving Students Across Achievement Definitions



(a) Average Grades 3 to 8 Reading Test



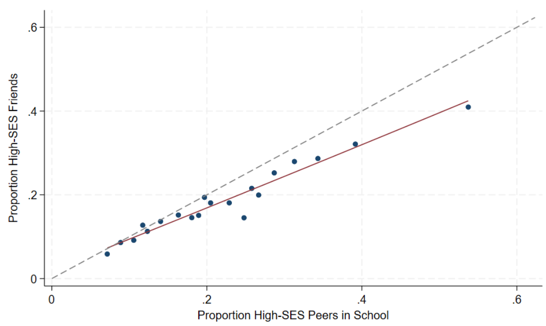
(b) Average Grades 3 to 8 Math Test



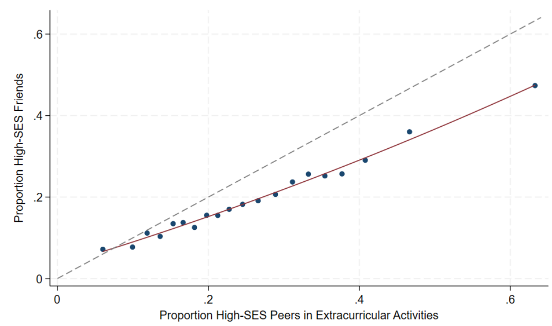
(c) Math G4 Test

Notes: Plots capture the average cumulative proportion of total classmates that are higher-achieving by student income. All tests are standardized within year-grade. High achieving students are those who are in the top 24 percentiles of the average grades 3 to 8 standardized tests (or grade 4 test in panel (c)). Cumulative exposure based on following three cohorts of grade 5 students (2012-2014) to expected grade 12.

Figure A8: Strong Relationship Between Students' School and Extracurricular Composition and Friendship Pattern



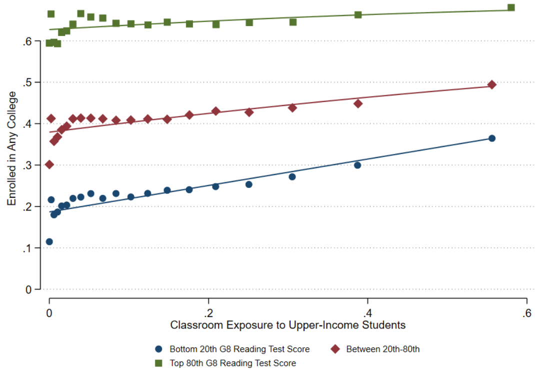
(a) Proportion of High-SES Peers in School



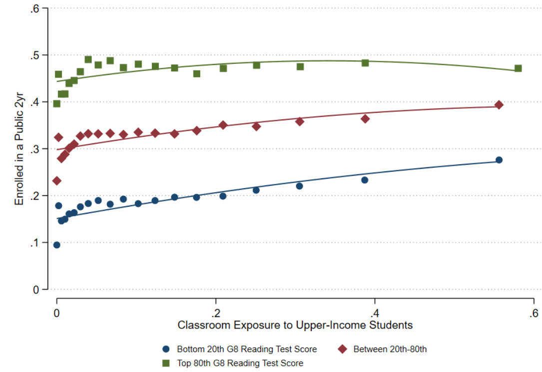
(b) Proportion of High-SES Peers in School Extracurricular Activities

Notes: The definition of high-SES is described in Section 3.2. The binned scatter plots are based on the full sample of low-SES students with data on friendship patterns. The bins are based on grouping the x-values into 20 equal-sized bins. Then, it computes the average y-variable value for each bin. The fitted lines are weighted by survey sampling weights. Panel (a) captures the relationship between a student's proportion of high-SES peers in school and their proportion of high-SES friends. Panel (b) captures the relationship between a student's proportion of high-SES peers in their school-listed extracurricular activities and their proportion of high-SES friends. In both panels, the dashed line is a 45-degree line representing a 1-to-1 relationship between the average proportion of high-SES students in school/extracurriculars and the average proportion of high-SES friends.

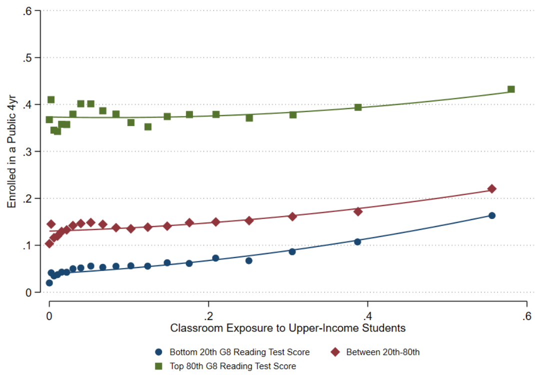
Figure A9: Exposure to Higher-Income Students, College Enrollment and Employment



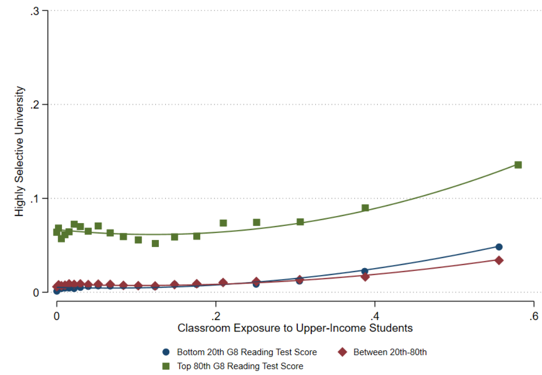
(a) Enrolled in Any College



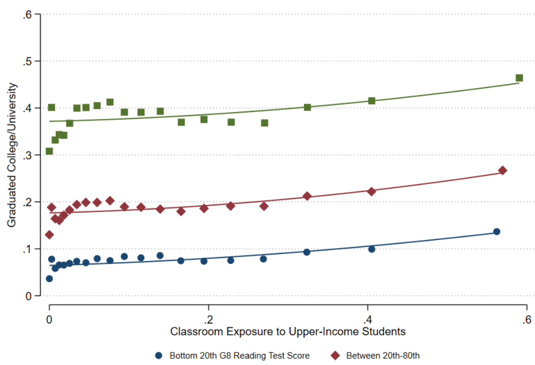
(b) Enrolled in 2-year Public College



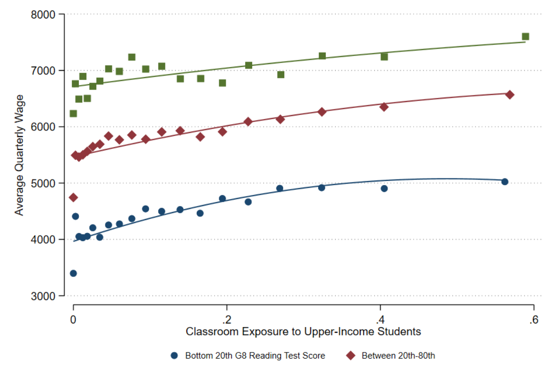
(c) Enrolled in 4-year Public University



(d) Enrolled in Highly Selective University



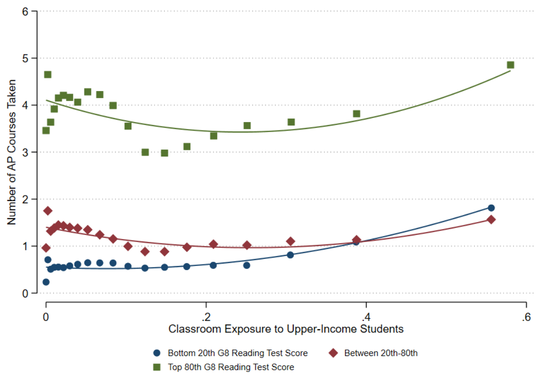
(e) Graduated from Any College/University



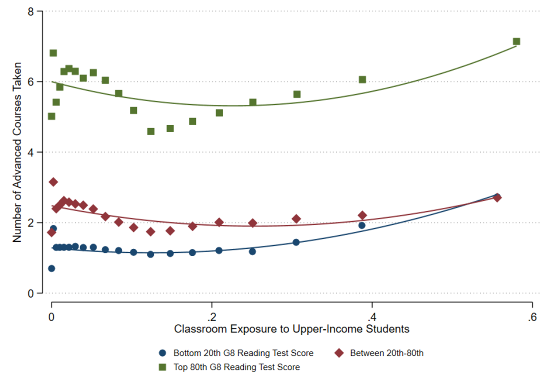
(f) Average Quarterly Wage in 2023 (Age 23-28)

Notes: The binned scatter plots are based on lower-income students and present the raw observational relationship (with no controls) between classroom exposure to upper-income students in high school and college enrollment and wages. Students are placed into three groups based on their standardized grade 8 reading test score: students who scored in the top 80th, between 20th and 80th, and 20th percentile of the cohort test score distribution. Test scores are standardized within the full sample of students who had taken the grade 8 test in a given year—92% of the students in the sample have grade 8 reading test scores. College enrollment outcomes include cohorts 2011 to 2019. College graduation and wage outcomes include cohorts 2011 to 2014. The bins are based on grouping the x-values into 20 equal-sized bins. Then, it computes the average y-variable value for each bin.

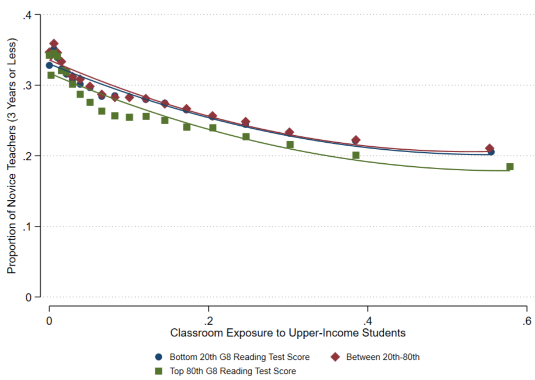
Figure A10: Exposure to Higher-Income Students and School Resources



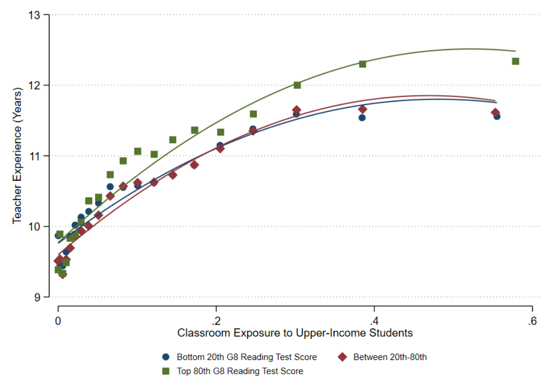
(a) Number of AP Courses Enrolled in



(b) Number of Advanced Courses Enrolled in



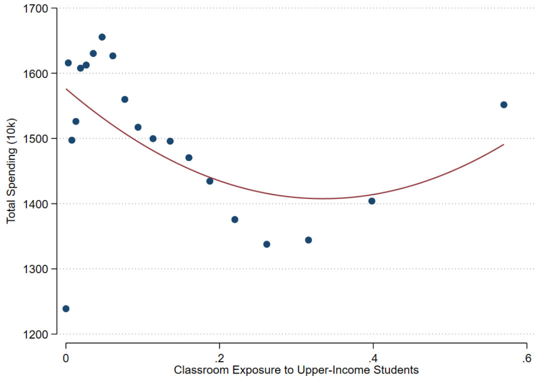
(c) Proportion of Novice Teachers



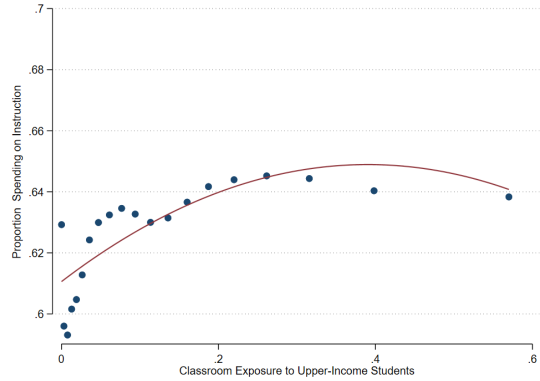
(d) Years of Teacher Experience

Notes: The line and plots are similar to those in Figure (A9). AP and advanced course enrollments are based on the average number of AP courses taken by the student during high school. The course enrollment outcomes include cohorts 2011 to 2019. The proportion of novice teachers is based on the proportion of full-time teachers with three or less years of experience that taught in a student's classroom in high school. Teacher data include cohorts 2012 to 2019.

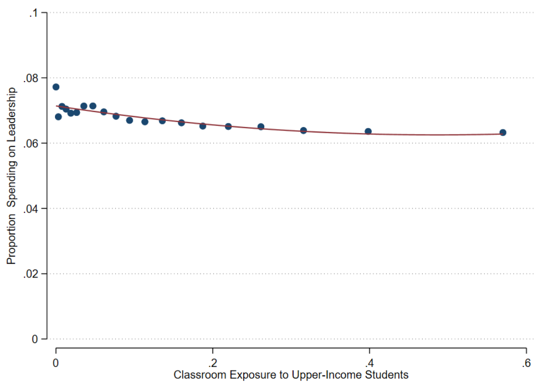
Figure A11: Exposure to Higher-Income Students and School Spending per Pupil



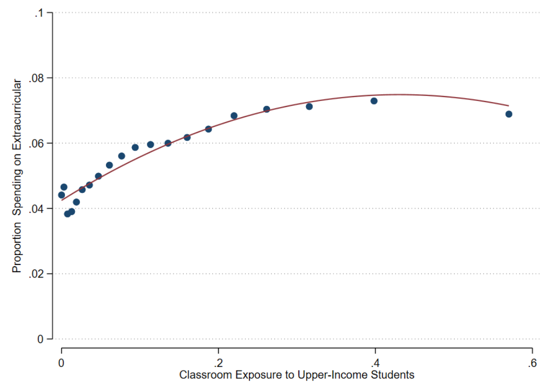
(a) Average School Spending per Pupil



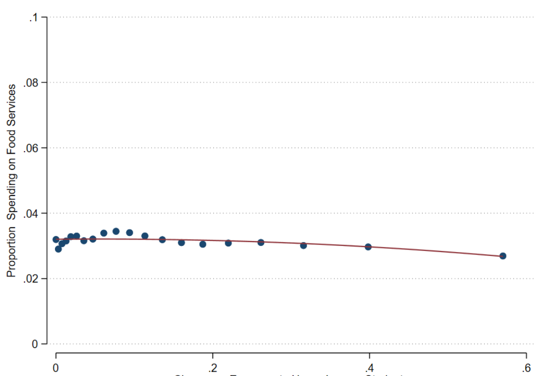
(b) Proportion of Total Spending on Instruction



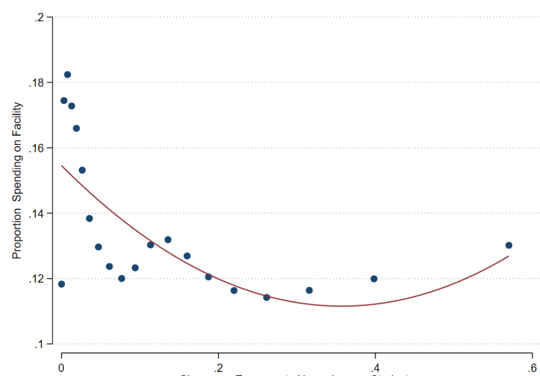
(c) Proportion of Total Spending on School Counselors



(d) Proportion of Total Spending on Co-/Extracurriculars



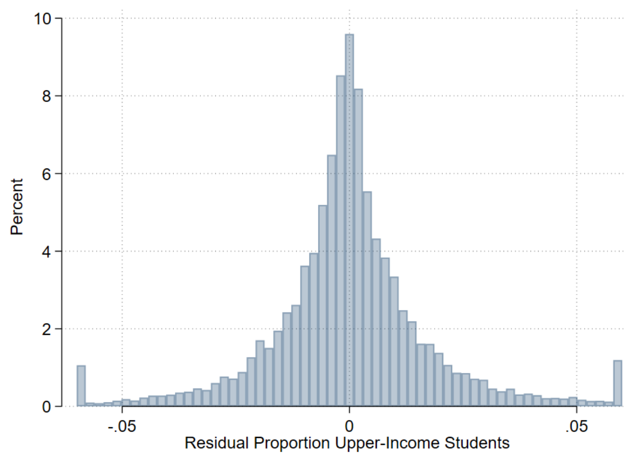
(e) Proportion of Total Spending on Food Services



(f) Proportion of Total Spending on Facilities

Notes: Spending is based on school actual spending data between 2012 and 2019 in each year divided by the total number of students enrolled in the school that year. Panel (a) captures the average yearly spending per pupil during the cohort's high school years. Panels (b)–(f) capture the proportion of total spending on each category. The instruction category combines the following functions: instruction, instruction resources and media services, curriculum and staff development, and instructional leadership. Facilities combines the following functions: facility acquisition and construction and facility maintenance/operations. With the exception of food services, I only included categories that make up more than 4% of the budget, on average. The data include cohorts 2012 to 2019. The bins are based on grouping the x-values into 20 equal-sized bins. Then, it computes the average y-variable value for each bin.

Figure A12: Distribution of Residual Variation in Proportion Upper-Income Students



Notes: The histogram presents the variation in the proportion of upper-income students based on the residual from a regression with school and cohort fixed effects, as well as school time trends.